PTC
多元微积分
习题册

PTC
Exercise book For Multivariable Calculus

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MTH004&MTH008

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在此，感谢各位对 PTC 伴学社的支持，同时，也感谢每一位参与习题册编写的成员所付出的努力

预祝各位阅读愉快，复习顺利，在期末考试中取得惊喜的成绩！

PTC 多元微积分资料组 全员——

董诗文　胡劭明　胡悦涵　江州　宋嘉晨
汤诗淼　滕启睿　杨涵淇　张佳妮　赵雨轩

（按拼音首字母排序，顺序不分先后）

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Part 1 章节习题汇总

Chapter 11 判断

1. If **u** and **v** are any two vectors, then $\|\mathbf{u}\cdot\mathbf{v}\| \geq \|\mathbf{u}\|\|\mathbf{v}\|$.

2. If **u + v** and **u - v** are perpendicular, then $\|\mathbf{u}\| = \|\mathbf{v}\|$.

3. For every vector **u**, $\|\mathbf{u}\|\cdot\mathbf{u} = \mathbf{u}\cdot\|\mathbf{u}\|$.

4. If **u** = a**i** + b**j** + c**k** is a unit vector, then a, b, and c are direction cosines for **u**.

5. The dot product for vectors satisfies the associative law.

6. Multiplying each component of a vector **v** by the scalar a multiplies the length of **v** by a.

7. If the vectors **a**, **b**, **c** are linearly dependent, then the volume of the parallelepiped formed by the edges of **a**, **b**, **c** is 0.

8. For every vector **u**, **u** × **u** = 0

9. If **a**, **b**, **c**, and **d** all lie in the same plane, then $(\mathbf{a}\times\mathbf{b})\times(\mathbf{c}\times\mathbf{d}) = \mathbf{0}$

10. The curves given by $\mathbf{r}_1(t) = \sin t\mathbf{i} + \cos t\mathbf{j} + t^3\mathbf{k}$ and $\mathbf{r}_2(t) = \sin t^3\mathbf{i} + \cos t^3\mathbf{j} + t^9\mathbf{k}$ for $0 \leq t \leq 1$ are identical.

11. The motion along the curves given by $\mathbf{r}_1(t) = \sin t\mathbf{i} + \cos t\mathbf{j} + t^3\mathbf{k}$ and $\mathbf{r}_2(t) = \sin t^3\mathbf{i} + \cos t^3\mathbf{j} + t^9\mathbf{k}$ for $0 \leq t \leq 1$ are identical.

12. Any line can be represented by both parametric equations and symmetric equations.

Chapter 11 简答

1. An object's position P changes so that its distance from (1, 2, -3) always equals its distance from (2, 3, 2). Find the equation of the plane on which P lies.

2. The solid spheres $(x-1)^2+(y-2)^2+(z-1)^2 \leq 4$ and $(x-2)^2+(y-4)^2+(z-3)^2 \leq 4$ intersect in a solid. Find its volume.

3. Let ABCD be a tetrahedron where M and N are the mid-points of the opposite edges AB and CD. Please express **MN** in terms of **AD** and **BC**.

4. Find the equation of the plane through (5. 1. -2) perpendicular to $n = \langle 2, 4, 3 \rangle$. Then find the cosine of the angle between this plane and the one with equation $3x - 4y + 7z = 5$.

5. There is a sphere whose center is (1, 2, -1) and radius is 2. Find the point $P \in 2x - 3y + z = 5$-plane closest to the sphere.

6. There are three vectors $\mathbf{u} = \langle 3, 2, 1 \rangle, \mathbf{v} = \langle 1, 1, 2 \rangle, \mathbf{w} = \langle 1, 3, 3 \rangle$
   (a) Find the area of the face determined by **u** and **v**.
   (b) Find the cosine of the angle between **u** and the plane containing the face determined by **v** and **w**.

7. Consider the curve $\mathbf{r}(t) = (t^2 + 2t + 3)\mathbf{i} + (-2t^2 + t + 2)\mathbf{j} + (5t^2 - 7)\mathbf{k}$. Show that this curve lies on a plane and find the equation of the plane.

8. Find parametric equations of the line through (1, - 2, 3) that is perpendicular to both the x-axis and the line $\frac{x-4}{2} = \frac{y-3}{-1} = \frac{z}{5}$

9. Find the distance between the two lines $x = 2 - t, y = 3 + 3t, z = 2t$ and $x = -1 + t, y = 2 - 3t, z = -1 - 2t$.

10. Find the distance between the skew lines $x = 2 - t, y = 3 + 4t, z = 2t$ and $x = -1 + t, y = 2, z = -1 + 2t$.

11. Find the parametric equations for the tangent line to the curve determined by $\mathbf{r}(t) = t\mathbf{i} + \frac{1}{2}t^2\mathbf{j} + \frac{1}{3}t^3\mathbf{k}$ at P(2) = (2, 2, $\frac{8}{3}$).

12. Find the equation of the surface that results when the curve z = 2y in the yz-plane is resolved about the z-axis.

Chapter 11 判断解析

1. If **u** and **v** are any two vectors, then $\|\mathbf{u}\cdot\mathbf{v}\| \geq \|\mathbf{u}\|\|\mathbf{v}\|$.【F】
【Solution】
$\|\mathbf{u}\cdot\mathbf{v}\| = \|\mathbf{u}\|\ \|\mathbf{v}\|\cos\theta, -1 \leq \cos\theta \leq 1$

2. If **u + v** and **u - v** are perpendicular, then $\|\mathbf{u}\| = \|\mathbf{v}\|$.【T】
【Solution】
点乘为 0 即可得，或看成对角线垂直的平行四边形为菱形

3. For every vector **u**, $\|\mathbf{u}\|\cdot\mathbf{u} = \mathbf{u}\cdot\|\mathbf{u}\|$.【F】
【Solution】
标量和向量点乘没有意义

4. If **u** = a**i** + b**j** + c**k** is a unit vector, then a, b, and c are direction cosines for **u**.
【T】
【Solution】
方向余弦为$\frac{a}{\|\mathbf{u}\|}$，$\frac{b}{\|\mathbf{u}\|}$，$\frac{c}{\|\mathbf{u}\|}$, $\|u\|=1$

5. The dot product for vectors satisfies the associative law.【F】
【Solution】
（**u**・**v**）・**w** 无意义

6. Multiplying each component of a vector **v** by the scalar a multiplies the length of **v** by a.【F】
【Solution】
长度为正，应为 v 的模长乘|a|

7. If the vectors **a**, **b**, **c** are linearly dependent, then the volume of the parallelepiped formed by the edges of **a**, **b**, **c** is 0.【T】
【Solution】
a，**b**，**c** 共面，$\mathbf{b}\times\mathbf{c}$与 **a** 垂直，即点乘为 0

8. For every vector **u**, **u** × **u** = 0【F】
【Solution】
叉乘结果是向量

9. If **a**, **b**, **c**, and **d** all lie in the same plane, then$(\mathbf{a}\times\mathbf{b})\times(\mathbf{c}\times\mathbf{d}) = \mathbf{0}$【T】
【Solution】
$\mathbf{a}\times\mathbf{b}$与$\mathbf{c}\times\mathbf{d}$共线，叉乘为 0

10. The curves given by $\mathbf{r}_1(t) = \sin t\mathbf{i} + \cos t\mathbf{j} + t^3\mathbf{k}$ and $\mathbf{r}_2(t) = \sin t^3\mathbf{i} + \cos t^3\mathbf{j} + t^9\mathbf{k}$ for $0 \le t \le 1$ are identical.【T】
【Solution】
参数化曲线形状与变量无关

11. The motion along the curves given by $\mathbf{r}_1(t) = \sin t\mathbf{i} + \cos t\mathbf{j} + t^3\mathbf{k}$ and $\mathbf{r}_2(t) = \sin t^3\mathbf{i} + \cos t^3\mathbf{j} + t^9\mathbf{k}$ for $0 \le t \le 1$ are identical.【F】
【Solution】
质点在同一时刻位置不一定一样。参数取值相同，函数值不同，不能看成相同的运动

12. Any line can be represented by both parametric equations and symmetric equations.【F】
【Solution】
对称式不一定，如 y = 1

Chapter 11 简答解析

1. An object's position P changes so that its distance from (1, 2, -3) always equals its distance from (2, 3, 2). Find the equation of the plane on which P lies.
【辅助理解】
通过等量关系找出 P 点轨迹，考查两点之间的距离公式
【Solution】

If P (x, y, z) denotes the moving point, $\sqrt{(x-1)^2 + (y-2)^2 + (z+3)^2} = \sqrt{(x-2)^2 + (y-3)^2 + (z-2)^2}$ which simplifies to $x + y + 5z = 3/2$, a plane.

2. The solid spheres $(x-1)^2 + (y-2)^2 + (z-1)^2 \le 4$ and $(x-2)^2 + (y-4)^2 + (z-3)^2 \le 4$ intersect in a solid. Find its volume.
【辅助理解】

本题为拓展题，可通过定积分的应用$\int_{r-h}^{r} \pi(\sqrt{r^2 - x^2})^2\, dx$ 求出球缺公式

【Solution】
Note that the volume of a segment of height h in a hemisphere of radius r is $\pi h^2\left[r - (\frac{h}{3})\right]$. The resulting solid is the union of two segments, one for each sphere. Since the two spheres have the same radius, each segment will have the same value for h. h is the radius minus half the distance between the centers of the two spheres.

$h = 2 - \frac{1}{2}\sqrt{(2-1)^2 + (4-2)^2 + (3-1)^2} = \frac{1}{2}$, V = 2 π h^2 [r-(h/3)] = $\frac{11}{12}\pi$

3. Let ABCD be a tetrahedron where M and N are the mid-points of the opposite edges AB and CD. Please express **MN** in terms of **AD** and **BC**.

【辅助理解】

注意 tetrahedron 意思为四面体，**MN** = **BN** − **BM** = $\frac{1}{2}\mathbf{BD}+\frac{1}{2}\mathbf{BC}-\frac{1}{2}\mathbf{BA}=$

$\frac{1}{2}$ $(\mathbf{AD}+\mathbf{BC})$

【Solution】

MN = $\frac{1}{2}$ (AD+BC)

4. Find the equation of the plane through (5. 1. -2) perpendicular to $\mathrm{n}=\langle 2,4,3\rangle$. Then find the cosine of the angle between this plane and the one with equation $3\mathrm{x}-4\mathrm{y}+7\mathrm{z}=5$.

【辅助理解】

求平面之间的夹角转化为求法向量之间的夹角

【Solution】

The first plane is $2\mathrm{x}+4\mathrm{y}+3\mathrm{z}=8$, and $\mathrm{m}=\langle 3,\ -4,\ 7\rangle$

Thus,$\cos\theta=\frac{\mathbf{m}\cdot\mathbf{n}}{\|\mathbf{m}\|\|\mathbf{n}\|}=\frac{6-16+21}{\sqrt{9+16+49}\sqrt{4+16+9}}$ = $\frac{11\sqrt{2146}}{2146}$

5. There is a sphere whose center is (1, 2, -1) and radius is 2. Find the point $\mathrm{P}\in 2\mathrm{x}-3\mathrm{y}+\mathrm{z}=5$-plane closest to the sphere.

【辅助理解】

先通过与半径的距离比较判断平面和球的位置关系，设点 P，当 CP 垂直于平面时距离最短，CP 即为法向量的倍数

【Solution】

$$\mathrm{d}=\frac{|2-6-1-5|}{\sqrt{4+9+1}}=\frac{10}{\sqrt{14}}>2$$

Suppose the point P is (a, b, c), $\mathbf{CP}=\langle \mathrm{a}-1,\mathrm{b}-2,\mathrm{c}+1\rangle=\mathrm{t}\mathbf{n}=\langle 2\mathrm{t},\ -3\mathrm{t},\ \mathrm{t}\rangle$

Additionally, $\|\mathrm{t}\mathbf{n}\|=\mathrm{L}$, $\mathrm{t}=\frac{5}{7}$. Then P is $(\frac{17}{7},\ -\frac{1}{7},\ -\frac{2}{7})$

6. There are three vectors $\mathbf{u}=\langle 3,2,1\rangle, \mathbf{v}=\langle 1,1,2\rangle, \mathbf{w}=\langle 1,3,3\rangle$

(a) Find the area of the face determined by **u** and **v**.

(b) Find the cosine of the angle between **u** and the plane containing the face determined by **v** and **w**.

【辅助理解】

两个向量叉乘的绝对值为围成的平行四边形面积，混合积的绝对值为围成的平行六面体的体积，直线与平面的余弦等于直线与法向量的正弦

【Solution】

(a) $\mathrm{A}=\|\mathbf{u}\times\mathbf{v}\|=\sqrt{35}$

(b) Let θ be the angle. Then θ is the complement of the smaller angle between **u** and **v**×**w**

$\sin\theta = \frac{|\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})|}{\|\mathbf{u}\|\|\mathbf{v}\times\mathbf{w}\|} = \frac{9}{14}$

7. Consider the curve $\mathbf{r}(t) = (t^2 + 2t + 3)\mathbf{i} + (-2t^2 + t + 2)\mathbf{j} + (5t^2 - 7)\mathbf{k}$. Show that this curve lies on a plane and find the equation of the plane.

【辅助理解】

所有的切线的方向向量与法向量垂直即可说明，再由点法式确定平面

【Solution】

$$\mathbf{r}'(t) = (2t + 2)\mathbf{i} + (-4t + 1)\mathbf{j} + 10t\mathbf{k}$$

Let $\mathbf{n} = \langle a, b, c\rangle$, then $(2t + 2)a + (-4t + 1)b + 10tc = 0$

$$(2a - 4b + 10c)t + 2a + b = 0$$

$\begin{cases} 2a - 4b + 10c = 0 \\ 2a + b = 0 \end{cases}$, Let a = 1, then $\mathbf{n} = \langle 1, -2, -1\rangle$

$\mathbf{r}(0) = \langle 3, 2, -7\rangle$

The plane is $(x - 3) - 2(y - 2) - (z + 7) = 0$

8. Find parametric equations of the line through (1, - 2, 3) that is perpendicular to both the x-axis and the line $\frac{x-4}{2} = \frac{y-3}{-1} = \frac{z}{5}$

【辅助理解】

点已知，叉乘得出方向向量即可

【Solution】

x=1, y=-2 + 5t, z=3 + t

$\mathbf{v} = \langle 2, -1, 5\rangle \times \langle 1, 0, 0\rangle = \langle 0, 5, 1\rangle$

9. Find the distance between the two lines $x = 2 - t, y = 3 + 3t, z = 2t$ and $x = -1 + t, y = 2 - 3t, z = -1 - 2t$.

【辅助理解】

求两条平行线之间的距离，先分别在两条曲线上随机找一个点，再求方向向量即可

【Solution】

$P = (2, 3, 0), Q = (-1, 2, -1), \mathbf{QP} = \langle 3, 1, 1\rangle, v = \langle -1, 3, 2\rangle$

$d = \frac{\|\mathbf{QP}\times\mathbf{v}\|}{\|\mathbf{v}\|} = \frac{5}{7}\sqrt{21}$

10. Find the distance between the skew lines $x = 2 - t, y = 3 + 4t, z = 2t$ and $x = -1 + t, y = 2, z = -1 + 2t$.

【辅助理解】

先叉乘求出法向量，两异面曲线距离为 PQ 在法向量上的标量投影

【Solution】

$\mathbf{n} = v1 \times v2 = \langle 8, 4, -4\rangle$ $P = (2, 3, 0), Q = (-1, 2, -1), QP = \langle 3, 1, 1\rangle$

d = $\frac{|\mathbf{QP}\cdot\mathbf{n}|}{\|\mathbf{n}\|} = \sqrt{6}$

11. Find the parametric equations for the tangent line to the curve determined by $\mathbf{r}(t) = t\mathbf{i} + \frac{1}{2}t^2\mathbf{j} + \frac{1}{3}t^3\mathbf{k}$ at P(2) = (2, 2, $\frac{8}{3}$).

【辅助理解】

如果把 r(t)看成位移，是关于时间的函数，那求导后的 v(t)速度就是位移轨迹的切线，所以求导后将 t=2 代入得到切线的方向向量，再结合切点得到直线的参数方程

【Solution】

x=2+t, y=2+2t, z=8/3+4t

12. Find the equation of the surface that results when the curve z = 2y in the yz-plane is resolved about the z-axis.

【辅助理解】

曲线绕哪个轴旋转，该变量不变，另一变量改为其与缺失变量平方和的正负平方根

【Solution】

At z = k, the revolution generates a circle of radius y = $\frac{z}{2}$ = $\frac{k}{2}$. Thus, the cross section in the plane z = k is the circle $x^2 + y^2 = \frac{k^2}{4}$, The equation of the surface is $z^2 = 4x^2 + 4y^2$.

Chapter 12 判断

1. If a function $f(x,y)$ has both partial derivatives at the origin $(0,0)$, then f is continuous at $(0,0)$.

2. For a function $f(x,y)$ with both mixed second-order partial derivatives f_{xy} and f_{yx} existing at a point (a,b), it must hold that $f_{xy}(a,b) = f_{yx}(a,b)$.

3. If f(x,y) has the same limit along every straight line approaching the origin (0,0), then $\lim_{(x,y)\to(0,0)} f(x,y)$ exists.

4. If f(x,0) is continuous at x=0 and f(0,y) is continuous at y=0, then f(x,y) is continuous at (0,0).

5. The direction of the gradient vector is the direction of fastest decrease of the function at that point.

6. The tangent plane to the surface F(x,y,z) = 0 at point (a,b,c) is F_x(a,b,c)(x−a) + F_y(a,b,c)(y−b) + F_z(a,b,c)(z−c) = 0.

7. The maximum value of the directional derivative equals the magnitude of the gradient vector.

8. If $f(x,y)$ is differentiable at (a,b), then the directional derivative exists in **any direction**.

9. if $z = f(x,y)$, $x = u^2 + v^2$, $y = 2uv$, then $\frac{\partial z}{\partial u} = 2u\frac{\partial f}{\partial x} + 2v\frac{\partial f}{\partial y}$

10. If $w = f(x,y)$, $x = g(t)$, $y = h(t)$, and $\frac{dw}{dt} = 0$ for all t, then $\nabla f = 0$

11. Given $F(x,y,z) = 0$ defining $z = z(x,y)$, then $\frac{\partial^2 z}{\partial x \partial y} = -\frac{F_{xy}F_z - F_xF_{zy}}{F_z^2}$

12. For any surface S defined implicitly by F(x, y, z) = 0 , if the tangent plane at point P ∈ S is parallel to the xy -plane, then the z -coordinate of P must be zero.

13. If f(x,y) is continuous on a **closed bounded set** S, then f attains a maximum value on S.

14. The function $f(x,y) = \sqrt[3]{x+y^4}$ has neither a global minimum nor a global maximum value.

15. If $\nabla f(p_0) = 0$, then f has an extreme value at p_0.

Chapter 12 简答

1. If $F(x) = \frac{2x-y}{xy}$, find $F_x(3,-2)$ and $F_y(3,-2)$

2. Find the slope of the tangent to the curve of intersection of the surface $3z = \sqrt{36-9x^2-4y^2}$ and the plane $x = 1$ at the point $(1,-2,\frac{\sqrt{11}}{3})$.

3. If $f(x,y,z) = e^{-xyz} - \ln(xy - z^2)$, find $f_x(x,y,z)$

4. Find the indicated limit or state that it does not exist.

a. $\lim\limits_{(x,y)\to(0,0)} \frac{x^2+y^2}{x^4-y^4}$. b. $\lim\limits_{(x,y)\to(0,0)} \frac{x^4-y^4}{x^2+y^2}$. c. $\lim\limits_{(x,y)\to(0,0)} \frac{\tan(x^2+y^2)}{x^2+y^2}$.

5. Describe the largest set S on which it is correct to say that $f(x,y) = \frac{x^2+3xy+y^2}{y-x^2}$ is continuous.

6. Let $f(x,y,z) = x^2 + 2y^2 + 3z^2$. Find the gradient vector at (1,−1,2) and write the equation of the tangent plane to the level surface at this point.

7. Does the surface $z = e^{x^2-y}$ at (0,1,1) have a tangent plane parallel to the plane 2x−y+z=4?

8. Prove that the tangent plane to the surface $xy + yz + zx$=3 at (1,1,1) is $x + y + z = 3$.

9. Let $f(x,y) = x^2 + 3xy - y^2$. Find:
(1) The gradient vector ∇f at (1,2);
(2) The directional derivative along **v**=(3,4)

10. Prove that the directional derivative of $f(x,y,z)$ at point P along direction **u** satisfies $D_u f = \nabla f \cdot u$, and show that the maximum directional derivative equals the gradient's magnitude.

11. Given the parametric surface S: $r(u,v) = (u^2 + v^2, u^3, v^3)$ with $u, v \in R$

(1) Find the equation of the tangent plane at point P(2,1,1)

(2) Prove that for any u,v, the intercept of the tangent plane on the x-axis is constant

12. Let the surface S be defined by $x^3 + y^3 + z^3 = 3xyz + 1$

(1) Find all points P(x,y,z) ∈ S such that:

①The tangent plane is parallel to the xy-plane ;

②The intercept of the tangent plane on the z-axis is 1.

(2) Prove that for any (x,y,z) ∈ S ,the tangent plane equation can be written as ax+by+ cz = 1 , and find expressions for coefficients a,b,c

13. Find the extrema of $f(x, y) = x^2y - 6y^2 - 3x^2$.

14. A rectangular box whose edges are parallel to the coordinate axes is inscribed in the ellipsoid $36x^2+4y^2 + 9z^2 = 36$. What is the greatest possible volume for such a box?

15. Use Lagrange multipliers to find the dimensions of the right circular cylinder with maximum volume if its surface area is 24π

Chapter 12 判断解析

1. If a function $f(x,y)$ has both partial derivatives at the origin $(0,0)$, then f is continuous at $(0,0)$.【F】

【辅助理解】

偏导数存在≠函数联系。偏导数仅关注沿坐标轴方向的变化率，而连续性要求所有路径的极限都一致

【Solution】

False. Consider the function $f(x,y)=\frac{xy}{x^2+y^2}$ for $(x,y)\neq(0,0)$ and $f(0,0)=0$.

The partial derivatives at (0,0) exist:

$$f_x(0,0)=\lim_{h\to 0}\frac{f(h,0)-f(0,0)}{h}=0,\ f_y(0,0)=0.$$

However, approaching (0,0) along $y=x$, we find:

$$\lim_{x\to 0}\frac{x\cdot x}{x^2+x^2}=\frac{1}{2}\neq f(0,0),$$

so f is not continuous at (0,0).

2. For a function $f(x,y)$ with both mixed second-order partial derivatives f_{xy} and f_{yx} existing at a point (a,b), it must hold that $f_{xy}(a,b)=f_{yx}(a,b)$.【F】

【辅助理解】

看到混合偏导可能就会下两者相等的结论，但它的前提是二阶偏导数需要连续，但这里没有定义函数。所以就不对了。

【Solution】

False. Let we find the special function:

$$f(x,y)=\begin{cases}\frac{xy(x^2-y^2)}{x^2+y^2} & (x,y)\neq(0,0)\\ 0 & (x,y)=(0,0)\end{cases}.$$

At (0,0), first-order partial derivatives :

$$f_x(0,y)=\lim_{h\to 0}\frac{f(h,y)-f(0,y)}{h}=\lim_{h\to 0}\frac{h^2y-y^3}{h^2+y^2}=-y$$

$$f_y(x,0)=\lim_{k\to 0}\frac{f(x,k)-f(x,0)}{k}=\lim_{k\to 0}\frac{x^3-xk^2}{x^2+k^2}=x$$

Then compute f_{xy} and f_{yx}, $f_{xy}(0,0)=-1$, $f_{yx}(0,0)=1$

So, $f_{xy}\neq f_{yx}$.

3. If f(x,y) has the same limit along every straight line approaching the origin (0,0), then $\lim_{(x,y)\to(0,0)} f(x,y)$ exists.【F】

【辅助理解】
可能误认为“所有直线路径的极限相同”等价于“极限存在”。实际上，多元函数极限要求所有可能路径（包括曲线路径）的极限一致。即使沿直线的极限相同，仍可能存在其他路径导致极限不同。
【Solution】

False. Consider $f(x,y)=\frac{x^2y}{x^4+y^2}$ for $(x,y)\neq(0,0)$.

Along straight lines $y=kx$:

$$\lim_{x\to 0}\frac{x^2\cdot(kx)}{x^4+(kx)^2}=\lim_{x\to 0}\frac{kx^3}{x^4+k^2x^2}=0.$$

Along the parabolic path $y=x^2$:

$$\lim_{x\to 0}\frac{x^2\cdot x^2}{x^4+(x^2)^2}=\lim_{x\to 0}\frac{x^4}{2x^4}=\frac{1}{2}.$$

Since the limit differs along different paths, $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist

4. If f(x,0) is continuous at x=0 and f(0,y) is continuous at y=0, then f(x,y) is continuous at (0,0).【F】
【辅助理解】
可能错误地将“沿坐标轴的连续性”等同于“整体连续性”。然而，连续性要求函数在所有方向（包括非坐标轴方向）的极限等于函数值。仅坐标轴连续无法保证整体连续。
【Solution】

False. Let $f(x,y)=\begin{cases}1 & \text{if } xy\neq 0\\ 0 & \text{if } xy=0\end{cases}$

Along the axes (x=0 or y=0):$f(x,0)=0, f(0,y)=0$, both continuous.
At (0,0):$f(0,0)=0$.
Along the line y=x(x≠0):

$$\lim_{x\to 0} f(x,x)=1\neq f(0,0).$$

Thus, $f(x,y)$ is not continuous at (0,0).

5. The direction of the gradient vector is the direction of fastest decrease of the function at that point.【F】
【Solution】
The gradient points in the direction of fastest increase; the negative gradient is the direction of fastest decrease.

6. The tangent plane to the surface F(x,y,z) = 0 at point (a,b,c) is F_x(a,b,c)(x−a) + F_y(a,b,c)(y−b) + F_z(a,b,c)(z−c) = 0.【T】

【辅助理解】
●想要寻找切平面需要三个条件：
a. 函数在该点可微

b. 该点的具体坐标
c. 该点的梯度

●**隐式曲面** F(x,y,z)=0: 切平面方程直接使用梯度分量（如原题）

显式函数 z=f(x,y): 切平面方程需转换为 $z-z_0= \nabla f(x_0,y_0)\cdot(x-x_0,y-y_0)$

【Solution】
The forms of tangent plane have two types:

○$f_x(x,y,z)(x-x_0)+f_y(x,y,z)(y-y_0)+f_z(x,y,z)(z-z_0)=0$

○$z = f(x_0,y_0)+\nabla f(x_0,y_0)\cdot(x-x_0,y-y_0)$

7. The maximum value of the directional derivative equals the magnitude of the gradient vector.【T】
【辅助理解】

方向导数与梯度的关系：$D_u f(p)=\nabla f(P)\cdot u$ 其中 u 是单位向量且 $u=ai+bj+ck$

【Solution】
The maximum directional derivative occurs in the gradient direction and equals $\|\nabla f\|$

8. If $f(x,y)$ is differentiable at (a,b), then the directional derivative exists in **any direction**.【T】
【Solution】
Differentiability guarantees the existence of all directional derivatives

9. if $z=f(x,y)$, $x=u^2+v^2$, $y=2uv$, then $\frac{\partial z}{\partial u}=2u\frac{\partial f}{\partial x}+2v\frac{\partial f}{\partial y}$【F】

【Solution】

The correct answer is: $\frac{\partial z}{\partial u}=2u\frac{\partial f}{\partial x}+2v\frac{\partial f}{\partial y}+\frac{\partial f}{\partial x}\frac{\partial x}{\partial v}+\frac{\partial f}{\partial y}\frac{\partial y}{\partial v}$

10. If $w=f(x,y)$, $x=g(t)$, $y=h(t)$, and $\frac{dw}{dt}=0$ for all t, then $\nabla f=0$【F】

【Solution】

It only states that the aspect of the simulance is orthogonal to the tangent direction of the curve , but the magnitude itself is not necessarily zero.

11. Given $F(x,y,z)=0$ defining $z=z(x,y)$, then $\frac{\partial^2 z}{\partial x\partial y}=-\frac{F_{xy}F_z-F_xF_{zy}}{F_z^2}$【F】

【Solution】
The correct expression for the second-order condensation partial derivative is：

$$\frac{\partial^2 z}{\partial x \partial y} = -\frac{F_{xy}F_z^2 - F_xF_{zy}F_z - F_yF_{xz}F_z + F_xF_yF_{zz}}{F_z^3}$$

12. For any surface S defined implicitly by F(x, y, z) = 0 , if the tangent plane at point P ∈ S is parallel to the xy -plane, then the z -coordinate of P must be zero. 【F】

【Solution】
The tangent plane at P is parallel to the xy -plane if and only if the normal vector ∇F has no z -component, i.e., F_z = 0. This condition depends only on the partial derivative F_z at P, not on the z -coordinate of P .

13. If f(x,y) is continuous on a **closed bounded set** S, then f attains a maximum value on S.【T】
【辅助理解】
通过极大极小值存在定理(Theorem A)可知.
【Solution】
If f is continuous on **a closed bounded set** S, then f attains both a (global)maximum value and a (global)minimum value there.

14. The function $f(x,y) = \sqrt[3]{x + y^4}$ has neither a global minimum nor a global maximum value.【T】
【辅助理解】
step1 分析函数基本性质,（外层函数立方根单调递增, 内层函数值域为实数集）
step2 通过极限分析其有界性
step3 再寻找临界点（critical point）（驻点, 奇异点）验证结果
【Solution】
Along the x-axis, f(x,0)→±∞ as x→±∞.

15. If $\nabla f(p_0) = 0$, then f has an extreme value at p_0.【F】

【Solution】
For the cylindrical surface f(x,y)=y^3, f(**p**)=**0** for every **p** on the x-axis, but f(**p**) is not an extreme value.

Chapter 12 简答解析

1. If $F(x,y)=\frac{2x-y}{xy}$, find $F_x(3,-2)$ and $F_y(3,-2)$

【辅助理解】这类题目考察求偏导和计算
【Solution】Find the partial derivative

$$F_x(x,y)=\frac{1}{x^2},\ F_y(x,y)=-\frac{2}{y^2},$$

Now, evaluate $F_x(3,-2)=\frac{1}{9}$, $F_y(3,-2)=-\frac{2}{4}=-\frac{1}{2}$

2. Find the slope of the tangent to the curve of intersection of the surface $3z=\sqrt{36-9x^2-4y^2}$ and the plane $x=1$ at the point $(1,-2,\frac{\sqrt{11}}{3})$.

【辅助理解】
这类题目考察多元函数中曲面与平面相交曲线切线斜率的求解，就像切西瓜一样，刀就是给定的平面，先带入已给平面得到函数，再用链式法则求导，这样可以方便一点，最后带入求斜率。
【Solution】
First, substitute x=1 into the surface equation:

Give the surface equation $3z=\sqrt{36-9-4y^2}=\sqrt{27-4y^2}$

$$z=\frac{1}{3}\sqrt{27-4y^2}$$

Next, find the derivative of z ,

$$\frac{dz}{dy}=\frac{1}{6}(27-4y^2)^{-\frac{1}{2}}\times(-8y)=-\frac{4y}{3\sqrt{27-4y^2}}$$

Then, evaluate the derivative at the given point $(1,-2,\frac{\sqrt{11}}{3})$,

$$\frac{dz}{dy}\Big|_{y=-2}=-\frac{4\times(-2)}{3\sqrt{27-4\times(-2)^2}}=\frac{8\sqrt{11}}{33}$$

3. If $f(x,y,z)=e^{-xyz}-\ln(xy-z^2)$, find $f_x(x,y,z)$

【辅助理解】
本类题目考察偏导，正常计算即可。变量偏多，容易算错
【Solution】

$f_x(x,y,z)=-yze^{-xyz}-\frac{y}{xy-z^2}$

4. Find the indicated limit or state that it does not exist.

a. $\lim\limits_{(x,y)\to(0,0)}\frac{x^2+y^2}{x^4-y^4}$.　　b. $\lim\limits_{(x,y)\to(0,0)}\frac{x^4-y^4}{x^2+y^2}$.　　c. $\lim\limits_{(x,y)\to(0,0)}\frac{\tan(x^2+y^2)}{x^2+y^2}$.

【Solution】

$\lim_{(x,y)\to(0,0)} \frac{x^2+y^2}{x^4-y^4}. = \lim_{(x,y)\to(0,0)} \frac{x^2+y^2}{(x^2+y^2)(x^2-y^2)} = \lim_{(x,y)\to(0,0)} \frac{1}{x^2-y^2}. (\text{when } (x,y) \neq (0,0)).$

Now, approach the origin along the line y=mx. Substitute y=mx into the function:

We have $\frac{1}{x^2-(mx)^2} = \frac{1}{x^2(1-m^2)}$, when (x,y)→(0,0), the value of the function depends on m.

If m=0, $\lim_{x\to 0} \frac{1}{x^2(1-0)} = \lim_{x\to 0} \frac{1}{x^2} = \infty$,

If m=1, $\lim_{x\to 0} \frac{1}{x^2(1-1)}$ and $\frac{1}{x^2(1-1)}$ is undefined.

So the limit does not exist.

b. $\lim_{(x,y)\to(0,0)} \frac{x^4-y^4}{x^2+y^2} = \lim_{(x,y)\to(0,0)} \frac{(x^2+y^2)(x^2-y^2)}{x^2+y^2} = \lim_{(x,y)\to(0,0)} (x^2-y^2)$.(when (x,y)≠(0,0))

=0

c. let t=x^2+y^2.As $(x,y) \to (0,0)$, then t→0

so the limit $\lim_{(x,y)\to(0,0)} \frac{\tan(x^2+y^2)}{x^2+y^2} = \lim_{t\to 0} \frac{\tan t}{t} = \lim_{t\to 0} \frac{\sin t}{\cos t \cdot t} = \lim_{t\to 0} \frac{\sin t}{t} \cdot \lim_{t\to 0} \frac{1}{\cos t}$ =1

5. Describe the largest set S on which it is correct to say that $f(x,y) = \frac{x^2+3xy+y^2}{y-x^2}$ is continuous.

【辅助理解】

本类题目考察连续性的定义，分母不为零才能使函数有定义且连续，这题中分母不为零即可。

【Solution】

The function can be defined except $y - x^2 = 0$.

So, we can write the answer, S={(x,y)ϵR^2:$y - x^2 \neq 0$}.

6. Let $f(x,y,z) = x^2 + 2y^2 + 3z^2$. Find the gradient vector at (1,−1,2) and write the equation of the tangent plane to the level surface at this point.

【Solution】

●Gradient: ∇f=(2x,4y,6z), substituting gives (2,−4,12).

●Tangent plane: 2(x−1)−4(y+1)+12(z−2)=0 or simplified to 2x−4y+12z=30

7. Does the surface $z = e^{x^2-y}$ at (0,1,1) have a tangent plane parallel to the plane 2x−y+z=4?

【Solution】

●Surface normal vector:$\left(2xe^{x^2-y}, -e^{x^2-y}, -1\right)$, at (0,1,1) : (0,−1,−1).

●Target plane normal: (2,−1,1). Parallel normals require proportionality, which fails here

8. Prove that the tangent plane to the surface $xy + yz + zx$=3 at (1,1,1) is $x + y + z = 3$.

【Solution】

●Let F(x,y,z)=xy+yz+zx−3. Gradient ∇F=(y+z,x+z,x+y), at (1,1,1): (2,2,2).
●Tangent plane: 2(x−1)+2(y−1)+2(z−1)=0, simplified to x+y+z=3.

9. Let $f(x, y) = x^2 + 3xy - y^2$. Find:
(1) The gradient vector ∇f at (1,2);
(2) The directional derivative along **v**=(3,4)

【Solution】

(1) ∇f=(2x+3y,3x−2y), substituting (1,2) gives (8,−1).

(2) Unit direction vector: **u**=(5/3, 5/4). Directional derivative: ∇f·**u**=8·5/3 + (−1)·5/4 = 4.

10. Prove that the directional derivative of $f(x, y, z)$ at point P along direction **u** satisfies $D_u f = \nabla f \cdot u$, and show that the maximum directional derivative equals the gradient's magnitude.

【Solution】

●By definition $D_u f = \lim_{h \to 0} \frac{f(P - h \cdot u) - f(P)}{h}$. Using differentiability, $D_u f = \nabla f \cdot u$.
●By the Cauchy-Schwarz inequality, ∇f·**u** ≤ ‖∇f‖·‖**u**‖. Equality holds when **u** aligns with ∇f, giving $D_u f$ = ‖∇f‖.

11. Given the parametric surface S: $r(u, v) = (u^2 + v^2, u^3, v^3)$ with $u, v \in R$

(1) Find the equation of the tangent plane at point P(2,1,1)

(2) Prove that for any u,v, the intercept of the tangent plane on the x-axis is constant

【Solution】

(1) Tangent Plane Equation

First. find parameters u, v corresponding to $P(2,1,\ 1)$: $\begin{cases} u^2 + v^2 = 2 \\ u^3 = 1 \Longrightarrow u = 1 \\ v^3 = 1 \Longrightarrow v = 1 \end{cases}$

Compute partial derivatives:

$$r_u = (2u, 3u^2, 0) = (2,3,0) \text{ at } (u,v) = (1,1)$$
$$r_v = (2v, 0, 3v^2) = (2,0,3) \text{ at } (u,v) = (1,1)$$

Normal vector: $n = r_u \times r_v = \begin{vmatrix} i & j & k \\ 2 & 3 & 0 \\ 2 & 0 & 3 \end{vmatrix} = (9, -6, -6)$

Tangent plane equation:

$$9(x-2) - 6(y-1) - 6(z-1) = 0 \Longrightarrow 9x - 6y - 6z = 6$$

(2) Constant x-Intercept

General tangent plane at (u,v) :

$$(3u^2)\left(x - (u^2 + v^2)\right) - 3u^3y - 3v^3z = 0$$

Simplify to intercept form:

$$x = \frac{3u^3y + 3v^3z + 3u^2(u^2+v^2)}{3u^2} = u^2 + v^2 + \frac{uy}{u} + \frac{vz}{u}$$

For $y = z = 0$, intercept $x = u^2 + v^2$.

But from parametric equations, $x = u^2 + v^2$, so intercept equals x-coordinate, which is constant for any u, v.

12. Let the surface S be defined by $x^3 + y^3 + z^3 = 3xyz + 1$

(1) Find all points P(x,y,z) ∈ S such that:

①The tangent plane is parallel to the xy-plane ;

②The intercept of the tangent plane on the z-axis is 1.

(2) Prove that for any (x,y,z) ∈ S ,the tangent plane equation can be written as ax+by+ cz = 1 , and find expressions for coefficients a,b,c

【Solution】

(1) Points on Surface

Define $F(x,y,z) = x^3 + y^3 + z^3 - 3xyz - 1$.

Gradient:

$$\nabla F = (3x^2 - 3yz, 3y^2 - 3xz, 3z^2 - 3xy)$$

Condition 1: Tangent plane parallel to xy-plane ⇒ ∇F

$$k = 0$$
$$3z^2 - 3xy = 0 \Longrightarrow z^2 = xy$$

Condition 2 : intercept $z = 1 \Longrightarrow \frac{1}{c} = 1 \Longrightarrow c = 1$.

From tangent plane equation:

$$\frac{x}{a}+\frac{y}{b}+z=1 \Longrightarrow a=\frac{F_x}{F_z}, b=\frac{F_y}{F_z}$$

Solve system:

$$\begin{cases} x^3+y^3+z^3=3xyz+1 \\ z^2=xy \\ z=1 \end{cases}$$

Solutions: $(1,1,1),(-1,-1,1)$.

(2)Coefficients a, b, c

Tangent plane equation: $(F_x)(x-x_0)+(F_y)(y-y_0)+(F_z)(z-z_0)=0$

Substitute $F_x=3x^2-3yz, F_y=3y^2-3xz, F_z=3z^2-3xy$:

$$(3x^2-3yz)x+(3y^2-3xz)y+(3z^2-3xy)z=1$$

Simplify:

$$3x^3+3y^3+3z^3-9xyz=3 \Longrightarrow x^3+y^3+z^3-3xyz=1$$

Coefficients:

$$a=\frac{x^2-yz}{z^2-xy}, b=\frac{y^2-xz}{z^2-xy}, c=1$$

13. Find the extrema of $f(x,y)=x^2y-6y^2-3x^2$.

【辅助理解】

step1 求梯度$\nabla f=0$ 的驻点(x_0,y_0)

step2 计算 Hessian 矩阵的行列式($D=f_{xx}f_{yy}-{f_{xy}}^2$)

step3 根据 D 和f_{xx}的正负性判断

$$\begin{cases} D>0, f_{xx}<0, \text{局部最大} \\ D>0, f_{xx}>0 \rightarrow \text{局部最小} \\ D<0 \rightarrow \text{鞍点, 不是极值} \\ D=0 \rightarrow \text{无效} \end{cases}$$

（也是求局部极值的方法）

【Solution】

$\nabla f(x,y)=(2xy-6x, x^2-12y)=(0,0)$

At(0,0) and $(\pm 6,3)$,

$$D=f_{xx}f_{yy}-{f_{xy}}^2=(2Y-6)(12)-(2x)^2=4(18-6y-x^2), f_{xx}=2(y-3)$$

At (0,0): D=72>0 and $f_{xx}<0$, so local maximum at (0,0).

At $(\pm 6,3)$: D<0, so $(\pm 6,3)$are saddle points.

14. A rectangular box whose edges are parallel to the coordinate axes is inscribed in the ellipsoid $36x^2+4y^2+9z^2=36$. What is the greatest possible volume for such a box?

【辅助理解】

在计算盒子体积时，容易因忽略第一卦限卦限顶点坐标(x, y, z)的和盒子边长 2x, 2y, 2z 的关系，计算体积时漏乘 8.

盒子内切于椭球 $36x^2+4y^2+9z^2=36$，等价为约束条件：$36x^2+4y^2+9z^2=36$.

【Solution】

Let (x,y,z)be the coordinates of the vertex of the box in the first octant. The edges of the box are 2x, 2y, and 2z.

The volume is, V=(2x)(2y)(2z)=8xyz

We need to maximize xyz under the constraint:

$36x^2+4y^2+9z^2=36$(x,y,z>0)

Define the Lagrangian function, $L(x,y,z,\lambda)=xyz-\lambda(36x^2+4y^2+9z^2-36)$

Set the partial derivatives to zero, $\begin{cases} \frac{\partial L}{\partial x}=yz-72\lambda x=0 \\ \frac{\partial L}{\partial y}=xz-8\lambda y=0 \\ \frac{\partial L}{\partial z}=xy-18\lambda z=0 \\ \frac{\partial L}{\partial \lambda}-\lambda(36x^2+4y^2+9z^2-36=0 \end{cases}$

From the first two equations, y=3x

From the first and third equations, z=2x

Substitute y=3x and z=2x into the ellipsoid equation,

So that, $y=\sqrt{3}$,$z=\frac{2}{\sqrt{3}}$, $V=8xyz=\frac{16}{\sqrt{3}}$

15. Use Lagrange multipliers to find the dimensions of the right circular cylinder with maximum volume if its surface area is 24π

【辅助理解】

通过拉格朗日乘数法，在表面积固定为 24π 的条件下，求圆柱体半径 r=2、高 h=4 时的最大体体积

【Solution】

Maximize $V(r,h)=\pi r^2 h$,

subject to $S(r,h)=2\pi r^2+2\pi rh-24\pi=0$

Define the Lagrangian function, $(2\pi rh,\pi r^2)=\lambda(4\pi r+2\pi h,2\pi r)$

$rh=\lambda(2r+h)$, $r=2\lambda$, $r^2+rh=12$

Critical point is (2,4).

So, radius r=2, Heigh h = 4

Chapter 13 判断

1. If the function $f(x,y)$ is continuous on all of R, then f is integrable over R

2. If f(x,y)≤g(x,y)over a rectangular region R, then $\iint_R f\,dA \le \iint_R g\,dA$

3. The integral $\int_0^1 \int_y^1 e^{x^2}\ dx\,dy$ can be swapped to$\int_0^1 \int_0^2 e^{x^2} dy\,dx$

4. If f(x,y)=g(x)h(y), then $\int_a^b \int_c^d f(x,y)\,dy\,dx$=[$\int_a^b g(x)\,dx$][$\int_c^d h(y)\,dy$]

5. If a double intergral turns out to have a negative value,we know that more than half of the solid is aboved the xy-plane

6. The region D is the enclosed area of $x \ge 0$, $y \ge 0$, $x + y \le 2$
   Thus:
   $$\iint_D |x+y-1|d\sigma = \int_0^1 dx \int_0^{1-x} (1-x-y)dy + \int_1^2 dx \int_{1-x}^{2-x} (x+y-1)\,dy$$

7. The region D is the enclosed area of $x^2 + y^2 \le 1$, $y \ge x^2 - 1$
   Thus:
   $$\iint_D (x^3 + y^3 + \sin x + \cos y)\ d\sigma = \iint_D \cos y\,d\sigma$$

8. Suppose D_n is the region of $n \le x + y \le n + 1$ in the first quadrant
   Due to the value of the series $S = \sum_{n=}^{\infty} \iint_{D_n} e^{-(x+y)}\,d\sigma$ is irrelevant to n ,
   S diverges.

9. In polar coordinates, the area element for double integrals is dA=drd θ

10. If the region S is an annulus 1⩽r⩽2, then θ must range from 0 to 2π

11. In polar coordinates, the Jacobian factor r can be omitted when computing surface area.

12. The surface area of the plane z=ax+by+c over region D equals the area of D multiplied by $\sqrt{a^2 + b^2 + 1}$

13. If $\delta(x,y) = k$ gives the density of a lamina at (x,y), the coordinates of the center of mass of the lamina do not involve k.

14. $\int_0^2 \int_0^{2\pi} \int_0^1 drdudz$ represents the volume of a right circular cylinder of radius 1 and height 2.

15. There are eight possible orders of integration for a triple iterated integral.

16. In spherical coordinates, when ρ is a constant, it represents a conical surface

17. When calculating the triple integral $\iiint_E \sqrt{x^2+z^2}dV$ for the region bounded by the paraboloid $y = x^2 + z^2$ and the plane $y = 4$, the lower limit of integration for z in cylindrical coordinates is 0.

Chapter 13 简答

1. Suppose the region D is ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} \le 1$

   Calculate

$$I = \iint_D \sqrt{1 - \frac{x^2}{a^2} + \frac{y^2}{b^2}}\ d\sigma$$

2. Suppose the region D is the area of $y \ge x^2$

   Calculate $I = \iint_D e^{-(x^2+y^2)} d\sigma$

   More necessary information: Duo to the region D is unbounded, there is an additional given function to help you solve this problem:

$$\int_0^{\pi} e^{-\frac{\sin^2\theta}{\cos^4\theta}} d\theta \approx 1.15$$

3. Evaluate the double integral $\iint_S (x^2 + y^2)\, dA$, where S is the region inside the cardioid r=2(1+$\cos\theta$) but outside the circle r=2.

4. Find the volume of the solid bounded by the two paraboloids $z = x^2 + y^2$ and the plane $z = 4 - x^2 - y^2$.

5. Find the area of the paraboloid z =$x^2 + y^2$ within $x^2 + y^2 \leq 1$

6. Find the mass of the solid between the spheres $x^2 + y^2 + z^2 = 1$ and $x^2 + y^2 + z^2$ =9 if the density is proportional to the distance from the origin.

7. Moment of inertia I_x about the x-axis of the solid bounded by the cylinder $y^2 + z^2 = 4$ and the planes x-y=0, x=0 and z=0 if the density $\delta(x, y, z) = z$.
(Hint: You will need to develop your own formula; slice, approximate, integrate.)

8. Calculate $\iiint_E \sqrt{x^2 + y^2} dV$, where E is the region bounded by the cylindrical surface $x^2 + y^2 = 1$the plane z = 0, and the plane z = 2

9. Find the volume of the upper - half region V bounded by the sphere $x^2 + y^2 + 2^2 = 4$ and the cone $z = \sqrt{x^2 + y^2}$

Chapter 13 判断解析

1. If the function $f(x,y)$ is continuous on R, then f is integrable over R【T】
【辅助理解】
根据可积性定理，如果 f 在一个封闭矩形区域有界，并且除了一些个别光滑曲线外，f 是连续的，那么 f 在 R 上可积。
【Solution】
True

2. If f(x,y)≤g(x,y)over a rectangular region R, then $\iint_R f\,dA \le \iint_R g\,dA$【T】

【辅助理解】

二重积分具有可比性定理，对所有(x,y)属于 R 若 f(x,y)≤g(x,y)，有$\iint_R f\,dA \le \iint_R g\,dA$

【Solution】
True

3. The integral $\int_0^1 \int_y^1 e^{x^2}\,dx\,dy$ can be swapped to$\int_0^1 \int_0^2 e^{x^2} dy\,dx$【F】

【辅助理解】

可通过画图得到积分范围，交换顺序后正确积分应为 $\int_0^1 \int_0^x e^{x^2} dy\,dx$

【Solution】
False

4. If f(x,y)=g(x)h(y), then $\int_a^b \int_c^d f(x,y)\,dy\,dx=[\int_a^b g(x)\,dx][\int_c^d h(y)\,dy]$【T】

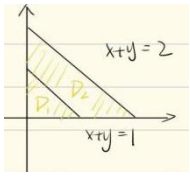
【辅助理解】
当 f(x,y)=g(x)h(y) 且积分区域为矩形时，二重积分可分解为两个单积分的乘积
【Solution】
True

5. If a double intergral turns out to have a negative value,we know that more than half of the solid is aboved the xy-plane【F】
【辅助理解】
如果二重积分的结果是负值，图像多的一半位于 xy 平面以下
【Solution】
False

6. The region D is the enclosed area of $x \ge 0$, $y \ge 0$, $x + y \le 2$ 【T】
Thus:

$$\iint_D |x+y-1|d\sigma = \int_0^1 dx \int_0^{1-x} (1-x-y)dy + \int_1^2 dx \int_{1-x}^{2-x} (x+y-1)\,dy$$

【辅助理解】



【Solution】

When $0 \le x + y \le 1$, the region D1: $\int_0^1 dx \int_0^{1-x} (1 - x - y) dy$

When $1 \le x + y \le 2$, the region D2: $\int_1^2 dx \int_{1-x}^{2-x} (x + y - 1)\, dy$

7. The region D is the enclosed area of $x^2 + y^2 \le 1$, $y \ge x^2 - 1$【F】
   Thus:

$$\iint_D (x^3 + y^3 + \sin x + \cos y)\ d\sigma = \iint_D \cos y\, d\sigma$$

【辅助理解】
所积区域不关于 x 对称，因此其它项无法忽略为 0，等式不成立
【Solution】
F

8. Suppose D_n is the region of $n \le x + y \le n + 1$ in the first quadrant
   Due to the value of the series $S = \sum_{n=}^{\infty} \iint_{D_n} e^{-(x+y)}\, d\sigma$ is irrelevant to n ,
   S diverges.【F】

【Solution】
Integral for x:

$$\iint_{D_n} e^{-(x+y)}\, d\sigma = \int_0^{n+1} dy \int_{n-y}^{\min(1,y)} e^{-(x+y)} dx$$

$$\int_{n-y}^{\min(1,y)} e^{-(x+y)} dx = -e^{-(x+y)} \Big|_{n-y}^{\min(1,y)} = -e^{-(\min(1,y)+y)} + e^{-(n-y+y)} = e^{-n} - e^{-(\min(1,y)+y)}$$

Then, integral for y:

$$\int_0^{n+1}\left(e^{-n}-e^{-(\min(1,y)+y)}\right)dy=\int_0^1\left(e^{-n}-e^{-(y+y)}\right)dy+\int_1^{n+1}\left(e^{-n}-e^{-(1+y)}\right)dy$$

$$=\int_0^1\left(e^{-n}-e^{-2y}\right)dy+\int_1^{n+1}\left(e^{-n}-e^{-(1+y)}\right)dy$$

$$=\left[ye^{-n}+\frac{1}{2}e^{-2y}\right]_0^1+\left[ye^{-n}+e^{-(1+y)}\right]_1^{n+1}$$

$$=e^{-n}+\frac{1}{2}e^{-2}-\frac{1}{2}+(n+1)e^{-n}+e^{-(n+2)}-e^{-n}-e^{-2}$$

$$=ne^{-n}+e^{-n}-\frac{1}{2}e^{-2}-\frac{1}{2}+e^{-(n+2)}=(n+1)e^{-n}-(n+2)e^{-(n+1)}$$

Thus,

$$S=\sum_{n=}^{\infty}\iint_{D_n}e^{-(x+y)}\,d\sigma \text{ is relevant to n.}$$

Besides,

$$S=\sum_{n=0}^{\infty}\left[(n+1)e^{-n}-(n+2)e^{-(n+1)}\right]$$

$$=\left(1\times e^{-0}-2\times e^{-1}\right)+\left(2\times e^{-1}-3\times e^{-2}\right)+\left(3\times e^{-2}-4\times e^{-3}\right)+\cdots$$

$$S=1-\lim_{n\to\infty}(n+2)e^{-(n+1)}$$

$$\lim_{n\to\infty}(n+2)e^{-(n+1)}=\lim_{n\to\infty}\frac{1}{e^{(n+1)}}=0$$

Therefore,

$$S=1\ ,\quad S\text{ converges}$$

9. In polar coordinates, the area element for double integrals is dA=drd θ 【F】

【辅助理解】

极坐标面积元素应该有 r

拓展："对于 dA=rdr dθ的理解"

①$dx\,dy=ds$

$ds=\frac{1}{2}d\theta\,(r+dr)^2-\frac{1}{2}d\theta\,r^2=\frac{1}{2}d\theta[2rdr+d^2r]=rdrd\theta+\frac{1}{2}d^2rd\theta$

($\frac{1}{2}d^2rd\theta$是无穷小）$ds\approx rdrd\theta$

②看成矩形（过于不严谨）

宽为 rdθ,长是 dr。$ds\approx rdrd\theta$

③叉乘（外积/楔形积）

dx与dy都是有方向，且两方向垂直，求面积则为叉乘

$dx\times dy=\ldots=[r\cos^2\theta+r\sin^2\theta]drd\theta$

【Solution】

False.

10. If the region S is an annulus 1⩽r⩽2, then θ must range from 0 to 2π【F】
【辅助理解】
θ的范围取决于具体区域定义，例如可能限制在第一象限。
【Solution】
False.

11. In polar coordinates, the Jacobian factor rr can be omitted when computing surface area.【F】
【辅助理解】
极坐标转换 dV=r dr dθ dz 需要包含雅可比因子 r
【Solution】
False

12. The surface area of the plane $z=ax+by+c$ over region D equals the area of D multiplied by $\sqrt{a^2+b^2+1}$【T】
【辅助理解】
平面偏导数 f_x=a, f_y=b，代入曲面面积公式化简得$\sqrt{a^2+b^2+1}$
【Solution】
True

13. If $\delta(x,y)=k$ gives the density of a lamina at (x,y), the coordinates of the center of mass of the lamina do not involve k.【T】
【辅助理解】
质心的坐标公式为$(\bar{x},\bar{y})=(\frac{M_y}{M},\frac{M_x}{M})$，其中总质量 $M=\iint\delta(x,y)dA$，静矩$M_y=\iint\delta(x,y)dA$和$M_x=\iint\delta(x,y)dA$。当 δ（x, y)=k 时，所有积分中 k 可提出，最终质心坐标中的 k 会被约去，因此质心坐标与 k 无关。
【Solution】
True

14. $\int_0^2\int_0^{2\pi}\int_0^1 drdudz$ represents the volume of a right circular cylinder of radius 1 and height 2.【F】
【辅助理解】
柱坐标系下的体积元素应为 $rdrd\theta dz$，而题目积分中缺少 r 因子。
【Solution】
False

15. There are eight possible orders of integration for a triple iterated integral.【F】

【Solution】
3!=6 种，False

16. In spherical coordinates, when ρ is a constant, it represents a conical surface【F】
【辅助理解】
在球面坐标里，ρ为常数时表示球面，φ为常数时才表示圆锥面，这里混淆了不同坐标变量对应常数时表示的曲面。
【Solution】
False

17. When calculating the triple integral $\iiint_E \sqrt{x^2+z^2}dV$ for the region bounded by the paraboloid $y = x^2 + z^2$ and the plane $y = 4$, the lower limit of integration for z in cylindrical coordinates is 0.【F】
【辅助理解】
在该区域中，z 的积分限需根据具体情况确定，并非一定是 0。$y = x^2 + z^2$和 $y = 4$，围成的区域，z 的范围是由抛物面决定的，不是固定从 0 开始
【Solution】
False

Chapter 13 简答解析

1. Suppose the region D is ellipse: $\frac{x^2}{a^2}+\frac{y^2}{b^2}\le 1$

   Calculate

$$I=\iint_D \sqrt{1-\frac{x^2}{a^2}+\frac{y^2}{b^2}}\ d\sigma$$

【Solution】

By using generalized polar coordinate transformation: $\begin{cases} x=ar\cos\theta \\ y=ar\sin\theta \end{cases}$

And Jacobian determinant:

$$J=\begin{vmatrix} a\cos\theta & -ar\sin\theta \\ b\sin\theta & br\cos\theta \end{vmatrix}=abr\ ,\ d\sigma=|J|\,dr d\theta=abr\,dr d\theta$$

The ranges of r and θ: $\begin{cases} 0\le r\le 1 \\ 0\le\theta\le 2\pi \end{cases}$

So

$$I=\int_0^{2\pi}d\theta\int_0^1\sqrt{1-r^2}\,abrdr=\frac{1}{2}ab\int_0^{2\pi}d\theta\int_0^1\sqrt{1-r^2}\,d\,r^2$$

$$=\frac{1}{2}ab\int_0^{2\pi}-\frac{2}{3}(1-r^2)^{\frac{2}{3}}\Big|_0^1 d\theta=\frac{1}{3}ab\int_0^{2\pi}1\,d\theta=\frac{2}{3}ab\pi$$

2. Suppose the region D is the area of $y\ge x^2$

   Calculate $I=\iint_D e^{-(x^2+y^2)}d\sigma$

   More necessary information: Duo to the region D is unbounded, there is an additional given function to help you solve this problem:

$$\int_0^{\pi} e^{-\frac{\sin^2\theta}{\cos^4\theta}}d\theta\approx 1.15$$

【Solution】

By using polar coordinate transformation: $\begin{cases} x=r\cos\theta \\ y=r\sin\theta \end{cases}$ $\quad d\sigma=r\,dr\,d\theta$

Duo to $y\ge x^2\ \rightarrow\ 0\le\ r\le\frac{\sin\theta}{\cos^2\theta}\quad ,\ \theta\in[0,\pi]$

$$\iint_D e^{-(x^2+y^2)}\,d\sigma=\int_0^{\pi}d\theta\int_0^{\frac{\sin\theta}{\cos^2\theta}}e^{-r^2}\cdot rdr$$

Integral for r，and let $u=r^2\ \rightarrow\ du=2rdr$

$$\int_0^{\frac{\sin^2\theta}{\cos^2\theta}} e^{-r^2}\cdot r dr = \frac{1}{2}\int_0^{\frac{\sin^2\theta}{\cos^2\theta}} e^{-u}du$$

$$= -\frac{1}{2}e^{-u}\Big|_0^{\frac{\sin^2\theta}{\cos^4\theta}} = -\frac{1}{2}\left(e^{-\frac{\sin^2\theta}{\cos^4\theta}} - 1\right)$$

$$= \frac{1}{2}\left(1 - e^{-\frac{\sin^2\theta}{\cos^4\theta}}\right)$$

Integral for θ,

$$\int_0^{\pi} \frac{1}{2}\left(1 - e^{-\frac{\sin^2\theta}{\cos^4\theta}}\right)d\theta$$

$$= \frac{1}{2}\int_0^{\pi} 1 d\theta - \frac{1}{2}\int_0^{\pi} e^{-\frac{\sin^2\theta}{\cos^4\theta}}d\theta$$

$$= \frac{\pi}{2} - \frac{1}{2}\int_0^{\pi} e^{-\frac{\sin^2\theta}{\cos^4\theta}}d\theta$$

Duo to

$$\int_0^{\pi} e^{-\frac{\sin^2\theta}{\cos^4\theta}}d\theta \approx 1.15$$

$$I \approx \frac{\pi}{2} - \frac{23}{40}$$

3. Evaluate the double integral $\iint_S (x^2 + y^2)\,dA$, where S is the region inside the cardioid r=2(1+$\cos\theta$) but outside the circle r=2.

【辅助理解】

该题考察极坐标下区域交点的确定、对称性简化积分和多项式展开积分

【Solution】

Analyze symmetry

The integrand in polar coordinates is:

$$\frac{x}{\sqrt{x^2 + y^2}} = \cos\theta$$

The region S is symmetric about the x-axis. However, $\cos\theta$ is an even function about $\theta = 0$, but the cardioid r=2(1+$\cos\theta$) is symmetric about the x-axis.

Identify integration limits

The region S exists between $\theta = -\frac{\pi}{2}$ and $\theta = \frac{\pi}{2}$. Due to the symmetry, integrate over $\theta \in [0, \frac{\pi}{2}]$.and multiply by 2:

$$\iint_S \cos\theta\, r\, \mathbf{dr}\, \mathbf{d}\theta = 2\int_0^{\frac{\pi}{2}}\int_0^{2(1+\cos\theta)} \cos\theta\, r\, \mathbf{dr}\, \mathbf{d}\theta$$

Integrate with Respect to r

$$\int_0^{2(1+\cos\theta)} r\, \mathbf{dr} = [\frac{r^2}{2}]_0^{1+\cos\theta} = 2(1+\cos\theta)^2$$

$$2\int_0^{\frac{\pi}{2}} \cos\theta\, 2(1+\cos\theta)^2\, \mathbf{d\theta} = 4\int_0^{\frac{\pi}{2}} \cos\theta\, (1+\cos\theta)^2\, \mathbf{d\theta}$$

$$4\int_0^{\frac{\pi}{2}} \cos\theta\, (1+2\cos\theta+\cos^2\theta)\mathrm{d}\theta = \frac{20}{3} + 2\pi\,.$$

4. Find the volume of the solid bounded by the two paraboloids $z = x^2 + y^2$ and the plane $z = 4 - x^2 - y^2$.

【辅助理解】

极坐标下立体投影区域的确定、上下曲面交线的求解

【Solution】

Set $x^2 + y^2 = 4 - x^2 - y^2$, which simplifies to $x^2 + y^2 = 2$. The projection region is a circle $r = \sqrt{2}$.

$$V = \iint_S ((4 - x^2 - y^2) - (x^2 + y^2))\, \mathbf{dA} = \iint_S ((4 - 2r^2)\, r\, \mathbf{dr}\, \mathbf{d}\theta$$

$$V = \int_0^{2\pi}\int_2^{\sqrt{2}} (4r - 2r^3)\, \mathbf{dr}\, \mathbf{d}\theta.$$

$$\int_0^{\sqrt{2}} (4r - 2r^3)\, \mathrm{dr} = [2r^2 - \frac{r^4}{2}]_0^{\sqrt{2}} = 4 - 2 = 2$$

$$V = \int_0^{2\pi} 2\, \mathrm{d}\theta = 4\pi$$

5. Find the area of the paraboloid $z = x^2 + y^2$ within $x^2 + y^2 \le 1$

【辅助理解】

运用极坐标代换解决计算问题目

【Solution】

The surface area formula :

A = $\iint_D \sqrt{(2x)^2 + (2y)^2 + 1}\mathrm{dA}$ = $\iint_D \sqrt{4\,(x^2 + y^2) + 1}\mathrm{dA}$

Let x=rcosθ, y=rsinθ, $0 \le r \le 1$, $0 \le \theta \le 2\pi$

Then A = $\int_0^{2\pi}\int_0^1 \sqrt{4r^2 + 1}\ r\, \mathrm{dr}\, \mathrm{d}\theta$ = $2\pi\int_0^1 \sqrt{4r^2 + 1}\ \mathrm{dr}$

Let u = $4r^2 + 1$

We have $\frac{\pi}{4}\int_1^5 \sqrt{u}\ \mathrm{du} = \frac{\pi}{6}(5\sqrt{5} - 1)$

6. Find the mass of the solid between the spheres $x^2+y^2+z^2=1$ and $x^2+y^2+z^2=9$ if the density is proportional to the distance from the origin.

【辅助理解】

坐标系选择：因问题具有球对称性，使用球坐标系简化积分。密度函数 δ（r)=kr（r 是到原点的距离）。

Tips:笛卡尔坐标下直接解析积分极其复杂，需转换为球坐标系简化计算。实际操作中，通过变量替换 x=rsinϕcosθ, y=rsinϕsinθ, z=rcosϕ，最终结果与球坐标系解法一致。

积分区域：介于半径 r=1 和 r=2 的球壳之间，角度φ∈[0, π]， θ ∈[0, 2π]。

分解积分：总质量是密度乘以体积元素的三重积分，通过分离变量逐项计算

【Solution】

$$M=\int_0^{2\pi}\int_0^{\pi}\int_1^3 kr\cdot r^2\sin\varphi dr d\varphi d\theta$$

$$=k\cdot\int_0^{2\pi}d\theta\cdot\int_0^{\pi}\sin\varphi d\varphi\cdot\int_1^3 r^3 dr$$

$$=k\cdot 2\pi\cdot 2\cdot\frac{80}{4}=80k\pi$$

7. Moment of inertia I_x about the x-axis of the solid bounded by the cylinder $y^2+z^2=4$ and the planes x-y=0, x=0 and z=0 if the density $\delta(x,y,z)=z$.

(Hint: You will need to develop your own formula; slice, approximate, integrate.)

【辅助理解】

坐标系选择：柱坐标系（y=rcosθ, z=rsinθ）, 因柱面 $y^2+z^2=4$ 对称。

积分区域：r∈[0, 2]（柱面半径）， θ ∈[0, 2π]（因 y≥0, z≥0），x∈[0, rcosθ]（由平面 x=y 和 x=0 限定）。（也能使用笛卡尔坐标计算，但过程更繁琐）

转动惯量公式：$I_x=\iiint(y^2+z^2)\delta dV$，其中$y^2+z^2=r^2$，密度$\delta=z=r\sin\theta$

简化积分：将体积元素替换为柱坐标 $dx\cdot rdr\cdot d\theta$, 逐层积分。

【Solution】

$$I_x=\int_0^{\frac{\pi}{2}}\int_0^2\int_0^{r\cos\theta} r^2\cdot r\sin\theta\cdot rdxdrd\theta.$$

$$=\int_0^{\frac{\pi}{2}}\int_0^2 r^5\sin\theta\cos\theta drd\theta$$

$$=\frac{1}{2}\cdot\frac{32}{3}=\frac{16}{3}$$

8. Calculate $\iiint_E\sqrt{x^2+y^2}dV$, where E is the region bounded by the cylindrical surface $x^2+y^2=1$the plane z = 0, and the plane z = 2

【Solution】

In cylindrical coordinates，x=r cosθ, y=r sinθ, dV=r dz dr dθ.

From $x^2+y^2=1$ we get $r^2=1$ so the range of r,θ, z are

$$\{(r,\theta,z)|0\le r\le 1, 0\le\theta\le 2\pi, 0\le z\le 2\}$$

$$\iiint_E \sqrt{x^2+y^2}dV = \int_0^{2\pi}\int_0^1\int_0^2 r^2\cdot rdzdr\,d\theta = \pi$$

9. Find the volume of the upper - half region V bounded by the sphere $x^2+y^2+2^2=4$ and the cone $z=\sqrt{x^2+y^2}$

【Solution】

In spherical coordinates $x=\rho\sin\varphi\cos\theta$, and $y=\rho\sin\varphi\sin\theta$, $z=\rho\cos\varphi$ and $dV=\rho^2\sin\varphi d\rho d_\theta d\varphi$.

From $x^2+y^2+2^2=4$, we have $\rho=2$

For the cone $z=\sqrt{x^2+y^2}$, $\tan\varphi=\frac{\sqrt{x^2+y^2}}{z}$ = 1, Since it's the upper - half region, the range of φ is $0\le\varphi\le\frac{\pi}{4}$

The range of θ is $0\le\theta\le 2\pi$

The volume V = $\int_0^{2\pi}\int_0^{\frac{\pi}{4}}\int_0^2 \rho^2\sin\varphi d\rho d_\theta d\varphi = \frac{8\pi(2-\sqrt{2})}{3}$

Chapter 9 判断

1. If a_n and b_n both diverge, then $a_n + b_n$ diverges.

2. If $\sum_{n=1}^{\infty} a_n$ converges, so does $\sum_{n=1}^{\infty} {a_n}^2$.

3. If for all natural numbers n $a_n > 0$ and $\sum_{n=1}^{\infty} a_n$ converges, them $\lim_{n\to\infty} (\frac{a_{n+1}}{a_n}) < 1$.

4. If for all natural numbers n, $0 \le a_{n+100} \le b_n$ and $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

5. If for all natural numbers n, $b_n \le a_n \le 0$ and if $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

6. If $\sum_{n=1}^{\infty} (u_{2n-1} + u_{2n})$ converges, then $\sum_{n=1}^{\infty} u_n$ must converge.

7. If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} \frac{a_n + a_{n+1}}{2}$ must converge.

8. If the series formed by grouping a positive series converges, then the original series may diverge.

9. A conditionally convergent series still converges after rearranging.

10. If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} |a_n|$ diverges.

11.If $0 \le a_n \le b_n$ for all natural numbers n and $\lim_{n\to\infty} b_n$ exists, then $\lim_{n\to\infty} a_n$ exists.

12. If $\lim_{n\to\infty} a_n = L$, then $\lim_{n\to\infty} a_{3n+4} = L$.

13. If $\lim_{n\to\infty} a_{2n} = L$, then $\lim_{n\to\infty} a_n = L$.

14.If a sequence $\{a_n\}$ converges to L, then the limit L is unique.

Chapter 9 简答

1. Determine if the series is convergent:$\sum_{n=1}^{\infty} \frac{\ln n}{n}$

2. Determine if the series is convergent:$\sum_{n=1}^{\infty} \frac{5n^2-4n}{2n+\sqrt{1+n^6}}$

3. Determine if this series is convergent$\sum_{n=2}^{\infty} (-1)^{n-1} \frac{n^2}{n^2-1}$.

4. $\sum_{n=1}^{\infty} \frac{\ln(1+\frac{1}{n^2})}{n^{\lambda}}$converges if and only if $\lambda >$ ( ).

5. Let a be a nonzero constant. Prove that $\sum_{n=1}^{\infty} \frac{a}{(-2)^n}$ is absolutely convergent.

6. Determine if the series is convergent: $\sum_{n=1}^{\infty} \frac{n!}{n^n}$.

7. Determine if the series is convergent: $\sum_{n=1}^{\infty} (-1)^n (1 - \cos\frac{a}{n})$, $(a > 0)$.

8. Calculate the sum of $\sum_{n=0}^{\infty} \frac{(\ln 3)^n}{2^n}$.

9. If $\sum_{n=1}^{\infty} \frac{(-1)^n + a}{n}$ converges, then determine the range of values for a.

10. Determine convergence or divergence for the series. $\sum_{n=1}^{\infty} \frac{n^n}{(2n)!}$

11. Determine convergence or divergence for the series. $\sum_{n=3}^{\infty} \frac{1}{[\ln(\ln n)]^{\ln n}}$

12. If $\sum_{n=2}^{\infty} \frac{\ln n!}{n^p}$ converges, find the range of values for P

13. Prove that $\sum_{n=1}^{\infty} (-1)^{n-1} \left(\frac{1}{2n} - \int_0^1 \frac{x^n}{1+x} dx\right)$ converges absolutely.

(Hint: prove $\frac{1}{2(n+1)} \le \int_0^1 \frac{x^n}{1+x} dx \le \frac{1}{2n}$)

14. Identify the sequences are convergent or divergent.

(1), $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \left(1 - \frac{1}{n}\right)$

(2), $\lim_{n\to\infty} b_n = \lim_{n\to\infty} \left(1 + (-1)^n \frac{1}{n}\right)$

(3), $\lim_{n\to\infty} c_n = \lim_{n\to\infty} \left[(-1)^n + \frac{1}{n}\right]$

(4), $\lim_{n\to\infty} d_n = 0.999$

15. Find $\lim_{n\to\infty}\frac{3n^2}{7n^2+1}$

16. Show that $\lim_{n\to\infty}\frac{\sin^3 n}{n}$ = 0

17. Write the first four terms of the sequence$\{a_n\}$, then use Theorem D to show that the sequence converges.

(1), $a_n=\frac{4n-3}{2^n}$

(2), $a_1=1$, $a_{n+1}=1+\frac{1}{2}a_n$

18. Find the convergence set of the power series $\Sigma_{n=1}^{\infty}\frac{(-1)^{n-1}}{n\cdot 3^n}x^n$

19. Find the sum function S(x) of the power series $\sum_{n=1}^{\infty} nx^{n-1}$and determine its convergence set.

20. Expand the function $f(x) = \frac{1}{(1-x)^2}$ as a power series in x and find its convergence set.

21. Find the limit of the function $\lim_{x\to 0} \frac{x^2 - x\ln(1+x)}{\sqrt{1+\tan x} - \sqrt{1+\sin x}}$

22. Find the convergence domain of the series,

$$\sum_{n=1}^{\infty} \left[1 - n\ln\left(1+\frac{1}{n}\right)\right] x^n$$

Chapter 9 判断解析

1. If a_n and b_n both diverge, then $a_n + b_n$ diverges.【F】

【辅助理解】

由$\sum u_n$和$\sum v_n$生成的级数$\sum(u_n + v_n)$满足收敛性：发散＋发散=无法判断，当且仅当两个级数均为正项才满足发散＋发散=发散，题目中没有给出a_n和b_n的限制。

2. If $\sum_{n=1}^{\infty} a_n$ converges, so does $\sum_{n=1}^{\infty} {a_n}^2$.【F】

【辅助理解】

如果正项级数$\sum u_n$收敛，那么$\sum {u_n}^2$和$\sum {u_n}^3$收敛，若不是正项级数则无法判断。

3. If for all natural numbers n $a_n > 0$ and $\sum_{n=1}^{\infty} a_n$ converges, them $\lim_{n\to\infty}(\frac{a_{n+1}}{a_n}) < 1$.【F】

【辅助理解】

Ratio Test 比率判别法不可逆转，正项级数$\lim_{n\to\infty}(\frac{a_{n+1}}{a_n}) < 1$ 才可推得$\sum_{n=1}^{\infty} a_n$收敛。

4. If for all natural numbers n, $0 \le a_{n+100} \le b_n$ and $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.【T】

【辅助理解】

比较判别法对所有的 $n < N$ 都成立，我们只关注 n 足够大的情况，有限项（a_n 的前 100 项）的差异不影响比较判别法的结果，因为$\sum_{n=1}^{\infty} b_n$收敛，所以$\sum_{n=101}^{\infty} a_n$（即$\sum_{n=1}^{\infty} a_{n+100}$）收敛，正项级数有部分收敛性可知，$\sum_{n=1}^{\infty} a_n$收敛。

5. If for all natural numbers n, $b_n \le a_n \le 0$ and if $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges. 【T】

【辅助理解】

比较判别法满足两个正项级数均大于 0，要化为 $0 \le -a_n \le -b_n$。因为$\sum_{n=1}^{\infty} b_n$收敛，所以$\sum_{n=1}^{\infty} -b_n$收敛，$-a_n \le -b_n$，所以$\sum_{n=1}^{\infty} -a_n$收敛，推出$\sum_{n=1}^{\infty} a_n$收敛。

6. If $\sum_{n=1}^{\infty}(u_{2n-1} + u_{2n})$ converges, then $\sum_{n=1}^{\infty} u_n$ must converge. 【F】

【辅助理解】

$\sum_{n=1}^{\infty}(u_{2n-1} + u_{2n})$是$\sum_{n=1}^{\infty} u_n$加括号后所得到的级数，前者不能推后者，但是后者能推前者（即题目的逆命题正确），反例：$\sum_{n=1}^{\infty}(-1)^{n-1}$.

7. If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} \frac{a_n+a_{n+1}}{2}$ must converge. 【T】

【辅助理解】

在级数中添加有限项不会影响级数的敛散性，所以当$\sum_{n=1}^{\infty} a_n$收敛，那么$\sum_{n=1}^{\infty} a_{n+1}$也收敛，再根据常数项级数的基本性质，相加或乘上标量后依旧是收敛的。

8. If the series formed by grouping a positive series converges, then the original series may diverge. 【F】

【辅助理解】

正项级数加括号所成的级数收敛，那么原级数也收敛，且和不变。

9. A conditionally convergent series still converges after rearranging. 【F】

【辅助理解】

绝对收敛的级数重排后，仍然收敛且和不变。但是条件收敛的级数重排后的新级数可能发散也可能收敛到任意有限的数。

10. If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} |a_n|$ diverges.【T】

【辅助理解】

$\sum_{n=1}^{\infty} a_n$ 发散可推出其绝对值也发散，但$\sum_{n=1}^{\infty} a_n$ 收敛不可推出其绝对值也收敛，要考虑绝对收敛和条件收敛的情况。

11. If $0 \le a_n \le b_n$ for all natural numbers n and $\lim_{n\to\infty} b_n$ exists, then $\lim_{n\to\infty} a_n$ exists.【F】

【辅助理解】

If $a_n=50+(-1)^n$, $b_n=100$,

Compare with the ordinary comparison test:

When $0 \le a_n \le b_n$, if $\sum b_n$ converges, so does a_n; if $\sum a_n$ diverges, so does b_n.

12. If $\lim_{n\to\infty} a_n = L$, then $\lim_{n\to\infty} a_{3n+4} = L$.【T】

【辅助理解】

When $n\to 0, 3n+4\to\infty$, then $\lim_{n\to\infty} a_{3n+4} = L$.

13. If $\lim_{n\to\infty} a_{2n} = L$, then $\lim_{n\to\infty} a_n = L$.【F】

【辅助理解】

Different from question 2,

$a_n=\begin{cases} 1 & \text{when n is a prime number(质数)} \\ 0 & \text{when n is a composit number(合数)} \end{cases}$,then $\lim_{n\to\infty} a_n$ does not exist.

14. If a sequence $\{a_n\}$ converges to L, then the limit L is unique.【T】

【辅助理解】

By the uniqueness of the limit of a convergent sequence.

Chapter 9 简答解析

1. Determine if the series is convergent:$\sum_{n=1}^{\infty}\frac{\ln n}{n}$

【辅助理解】

我们使用比较判别法，把$\frac{\ln n}{n}$ 和 $\frac{1}{n}$对比进行放缩，转化为常数加上一个发散的调和级数的形式，进行比较。

【Solution】

When $n \geq 3, \ln n \geq 1$,so we compare $\frac{\ln n}{n}$ with $\frac{1}{n}$

$$\sum_{n=1}^{\infty}\frac{\ln n}{n}=\frac{\ln 1}{1}+\frac{\ln 2}{2}+\sum_{n=3}^{\infty}\frac{\ln n}{n}>\frac{\ln 1}{1}+\frac{\ln 2}{2}+\sum_{n=3}^{\infty}\frac{1}{n}=\infty$$

Since it includes the divergent harmonic series starting from n=3. Therefore, the series is divergent.

2. Determine if the series is convergent: $\sum_{n=1}^{\infty}\frac{5n^2-4n}{2n+\sqrt{1+n^6}}$

【辅助理解】

多项式除多项式的级数，我们需要先判断上下多项式最高次幂大小，如题中上多项式是二次幂，下式是三次幂，用极限比较判别法构造b_n使得上下式幂齐次，再同除以最高次幂正常求极限。

【Solution】

Let $a_n=\frac{5n^2-4n}{2n+\sqrt{1+n^6}}$, $b_n=\frac{1}{n}$

$$\lim_{n\to\infty}\frac{a_n}{b_n}=\lim_{n\to\infty}\frac{\frac{5n^2-4n}{2n+\sqrt{1+n^6}}}{\frac{1}{n}}=\lim_{n\to\infty}\frac{n(5n^2-4n)}{2n+\sqrt{1+n^6}}=5>0$$

Since$\sum b_n$diverges, $\sum a_n$diverges as well.

3. Determine if this series is convergent$\sum_{n=2}^{\infty}(-1)^{n-1}\frac{n^2}{n^2-1}$.

【辅助理解】

判断交错级数$(-1)^n a_n$，要验证$a_n > a_{n+1} > 0, \lim_{n\to\infty} a_n = 0$ 即可.

前者需要求导，若导数小于 0，则a_n递减，满足$a_n > a_{n+1}$，再验证$\lim_{n\to\infty} a_n = 0$ 即可，切记注意两个条件要同时满足否则不成立。莱布尼茨判别法对交错级数的判断是单独成立的，比如$a_n = \frac{1}{n}$时，莱布尼茨判别法满足$\frac{1}{n} > \frac{1}{n+1} > 0$ 和$\lim_{n\to\infty}\frac{1}{n} = 0$，该交错级数收敛，但单独的调和级数$\sum\frac{1}{n}$是发散的。

【Solution】

This series is alternating, and decreasing because

$f(x)=\frac{x^2}{x^2-1}, f'(x)=-\frac{2x}{(x^2-1)^2}<0$ for $x>1$

However $\lim_{n\to\infty}\frac{n^2}{n^2-1}=1\neq 0$,so it is not convergent.

4. $\sum_{n=1}^{\infty}\frac{\ln(1+\frac{1}{n^2})}{n^{\lambda}}$converges if and only if $\lambda>(\ )$.

【辅助理解】

我们构造b_n，使用比较判别法，凑等价无穷小（由$\ln(1+\frac{1}{n^2})$中$\frac{1}{n^2}$趋于 0 推得：

要构造的b_n可以使极限约成$\frac{\ln(1+\frac{1}{n^2})}{\frac{1}{n^2}}$。然后可以得出$\lambda$的取值范围。

【Solution】

Let $a_n=\frac{\ln(1+\frac{1}{n^2})}{n^{\lambda}}$, $b_n=\frac{1}{n^{\lambda+2}}$

$$\lim_{n\to\infty}\frac{\frac{\ln(1+\frac{1}{n^2})}{n^{\lambda}}}{\frac{1}{n^{\lambda+2}}}=\lim_{n\to\infty}\frac{\ln(1+\frac{1}{n^2})}{\frac{1}{n^2}}=1$$

By Limit comparison test ,we should find that $\sum\frac{1}{n^{\lambda+2}}$ should be convergent. Therefore, $\lambda+2>1, \lambda>-1$.

5. Let a be a nonzero constant. Prove that $\sum_{n=1}^{\infty}\frac{a}{(-2)^n}$ is absolutely convergent.

【辅助理解】

根据绝对收敛的定义，$\sum_{n=1}^{\infty}u_n$的情况下如果$\sum_{n=1}^{\infty}|u_n|$也收敛，那么$\sum_{n=1}^{\infty}u_n$绝对收敛。把原级数化成几何级数的形式，分别判断$\sum_{n=1}^{\infty}u_n$和$\sum_{n=1}^{\infty}|u_n|$的收敛性。

【Solution】

We can rewrite $\sum_{n=1}^{\infty}\frac{a}{(-2)^n}$ as $\sum_{n=1}^{\infty}a\cdot(-\frac{1}{2})^n$.This is a geometric-series with the general term$u_n=a\cdot(-\frac{1}{2})^n$.

Let m=n-1, $\sum_{n=1}^{\infty}a\cdot(-\frac{1}{2})^n=\sum_{m=0}^{\infty}a\cdot(-\frac{1}{2})^{m+1}=-\frac{a}{2}\sum_{m=1}^{\infty}1\cdot(-\frac{1}{2})^{m-1}$

The common-ratio $|r|=\frac{1}{2}<1$.So $\sum_{n=1}^{\infty}\frac{a}{(-2)^n}$ is convergent.

Then, consider the absolute-convergence:

$\sum_{n=1}^{\infty}\frac{|a|}{(-2)^n}=\sum_{n=1}^{\infty}\frac{|a|}{2}\cdot(-\frac{1}{2})^{n-1}$.For the geometric-series $\sum_{n=1}^{\infty}\frac{|a|}{(-2)^n}$, $|r|=\frac{1}{2}<1$.

So $\sum_{n=1}^{\infty}\frac{|a|}{(-2)^n}$ is convergent.

Therefore, $\sum_{n=1}^{\infty}\frac{a}{(-2)^n}$ is absolutely convergent.

6. Determine if the series is convergent: $\sum_{n=1}^{\infty}\frac{n!}{n^n}$.

【辅助理解】

一般出现 n 阶乘或 n 次方，可以考虑使用正项级数的 Ratio Test

$$\lim_{n\to\infty}\frac{u_{n+1}}{u_n}=\lim_{n\to\infty}\frac{(n+1)!}{(n+1)^{n+1}}\cdot\frac{n!}{n^n}=\lim_{n\to\infty}\frac{1}{(1+\frac{1}{n})^n}=\frac{1}{e}<1$$

Hence the series is convergent by ratio test.

【Solution】Converge

7. Determine if the series is convergent: $\sum_{n=1}^{\infty}(-1)^n(1-\cos\frac{a}{n}),\ (a>0)$.

【辅助理解】

方法不唯一，交错级数可以用莱布尼茨判别法判断敛散性（求导判断单调性，验证通项趋于 0），还可以联想到等价无穷小，构造b_n，通过比较判别法判断。

Since $\lim_{n\to\infty}\frac{1-\cos\frac{a}{n}}{\frac{1}{2}(\frac{a}{n})^2}=1$, and $\sum_{n=1}^{\infty}\frac{a^2}{2n^2}$ converges, then $\sum_{n=1}^{\infty}(-1)^n(1-\cos\frac{a}{n})$ is absolutely convergent.

【Solution】Converge

8. Calculate the sum of $\sum_{n=0}^{\infty}\frac{(\ln3)^n}{2^n}$.

【辅助理解】

观察到是 Geometric Series，那么用等比级数求和公式直接秒了。

$$q=\frac{\ln3}{2},\ S=\frac{1}{1-\frac{\ln3}{2}}=\frac{2}{2-\ln3}$$

【Solution】$\frac{2}{2-\ln3}$

9. If $\sum_{n=1}^{\infty}\frac{(-1)^n+a}{n}$ converges, then determine the range of values for a.

【辅助理解】

基于课堂所学，$\sum_{n=1}^{\infty}\frac{(-1)^n}{n}$是收敛的，而$\sum_{n=1}^{\infty}\frac{1}{n}$是发散的，将我们已知的条件与题目结合起来！注意取值范围也可以是一个值！

Since $\sum_{n=1}^{\infty}\frac{(-1)^n}{n}$ and $\sum_{n=1}^{\infty}\frac{(-1)^n+a}{n}$ converges, then $\sum_{n=1}^{\infty}\frac{(-1)^n+a}{n}-\sum_{n=1}^{\infty}\frac{(-1)^n}{n}=\sum_{n=1}^{\infty}\frac{a}{n}$ also converges.

Thus, a must be equal to 0; otherwise, it diverges

【Solution】0

10. Determine convergence or divergence for the series. $\sum_{n=1}^{\infty}\frac{n^n}{(2n)!}$

【辅助理解】

$$\lim_{n\to\infty}\frac{\frac{(n+1)^{n+1}}{(2n+2)!}}{\frac{n^n}{(2n)!}}=\lim_{n\to\infty}\frac{n+1}{(2n+2)(2n+1)}=0$$

【Solution】converge

11. Determine convergence or divergence for the series. $\sum_{n=3}^{\infty}\frac{1}{[\ln(\ln n)]^{\ln n}}$

【辅助理解】

$$a_n=\frac{1}{[\ln(\ln n)]^{\ln n}}=\frac{1}{e^{\ln n\cdot\ln[\ln(\ln n)]}}=\frac{1}{n^{\ln[\ln(\ln n)]}}$$

当 n 足够大时，ln[ln(lnn)]>1 converge

【Solution】converge

12. If $\sum_{n=2}^{\infty}\frac{\ln n!}{n^p}$ converges, find the range of values for P

【辅助理解】

常数

$$\frac{\ln n!}{n^p}=\frac{\ln\sqrt[n]{n!}}{n^{p-1}}=\frac{(\ln\sqrt[n]{n!}-\ln n)+\ln n}{n^{p-1}}=\frac{\ln\frac{\sqrt[n]{n!}}{n}+\ln n}{n^{p-1}}\sim\frac{\ln n}{n^{p-1}}$$

$$\lim_{n\to\infty}\frac{\sqrt[n]{n!}}{n}=\lim_{n\to\infty}\sqrt[n]{\frac{n!}{n^n}}=\lim_{n\to\infty}\left(\frac{1\times2\times3\cdots n}{n\cdot n\cdot n\cdots n}\right)^{\frac{1}{n}}$$

$$=e^{\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\ln\frac{k}{n}}=e^{\int_0^1\ln x dx}=\frac{1}{e}$$

积分定义

$$\int_0^1 \ln x dx=[x\ln x]_0^1-\int_0^1 x\cdot\frac{1}{x}dx=-1$$

$p-1>1 \quad p>2$

$\therefore p\in(2,+\infty)$

【Solution】$p\in(2,+\infty)$

13. Prove that $\sum_{n=1}^{\infty}(-1)^{n-1}\left(\frac{1}{2n}-\int_0^1\frac{x^n}{1+x}dx\right)$ converges absolutely.

(Hint: prove $\frac{1}{2(n+1)}\le\int_0^1\frac{x^n}{1+x}dx\le\frac{1}{2n}$)

【Solution】

$$\int_0^1 \frac{1}{1+x}x^n dx \geq a_n = \int_0^1 \frac{1}{1+1}x^n dx = \frac{1}{2}a_n = \int_0^1 x^n dx = \frac{1}{2(n+1)}$$

$$\int_0^1 \frac{x}{1+x}x^{n-1}dx = a_n = \int_0^1 (1-\frac{1}{1+x})x^{n-1}dx \leq a_n = \int_0^1 \frac{1}{2}x^{n-1}dx = \frac{1}{2n}$$

$$\frac{1}{2(n+1)} \leq \int_0^1 \frac{x^n}{1+x}dx \leq \frac{1}{2n}$$

$0 \leq \frac{1}{2n} - \int_0^1 \frac{x^n}{1+x}dx \leq \frac{1}{2n} - \frac{1}{2(n+1)} = \frac{1}{2}\cdot\frac{1}{n(n+1)} \sim \frac{1}{n^2}$ converges

$\sum_{n=1}^{\infty} \frac{1}{2n} - \int_0^1 \frac{x^n}{1+x}dx$ converges,

so $\sum_{n=1}^{\infty} (-1)^{n-1} (\frac{1}{2n} - \int_0^1 \frac{x^n}{1+x}dx)$ converges absolutely

14. Identify the sequences are convergent or divergent.

(1), $\lim_{n\to\infty} a_n = \lim_{n\to\infty} (1-\frac{1}{n})$

(2), $\lim_{n\to\infty} b_n = \lim_{n\to\infty} (1+(-1)^n\frac{1}{n})$

(3), $\lim_{n\to\infty} c_n = \lim_{n\to\infty} [(-1)^n + \frac{1}{n}]$

(4), $\lim_{n\to\infty} d_n$ =0.999

【Solution】

(1),Limit =1, so it converges to 1

(2),Limit =1, so it converges to 1

(3),Limit does not exist, so it diverges

(4),Limit =0.999, it converges to 0.999

15. Find $\lim_{n\to\infty} \frac{3n^2}{7n^2+1}$

【Solution】

$$\lim_{n\to\infty} \frac{3n^2}{7n^2+1} = \lim_{n\to\infty} \frac{3}{7+\frac{1}{n^2}} = \frac{\lim_{n\to\infty} 3}{\lim_{n\to\infty} 7 + \lim_{n\to\infty} \frac{1}{n^2}} = \frac{3}{7+\lim_{n\to\infty} \frac{1}{n^2}} = \frac{3}{7}$$

16. Show that $\lim_{n\to\infty} \frac{\sin^3 n}{n}$ = 0

【Solution】

For n≥ 1, $-\frac{1}{n} \leq \frac{\sin^3 n}{n} \leq \frac{1}{n}$

Since $\lim_{n\to\infty} (-\frac{1}{n})$=0 and $\lim_{n\to\infty} \frac{1}{n}$=0

The result follows by the Squeeze Theorem

17. Write the first four terms of the sequence$\{a_n\}$, then use Theorem D to show that the sequence converges.

(1), $a_n = \frac{4n-3}{2^n}$

(2), $a_1 = 1$, $a_{n+1} = 1 + \frac{1}{2}a_n$

【Solution】

(1),First four terms:$a_1 = 0.5, a_2 = 1.25, a_3 = 1.125, a_4 \approx 0.8125$

For all $n \geq 1, a_n > 0$,

$a_n - a_{n+1} = \frac{8n-6}{2^{n+1}} - \frac{4n-3}{2^{n+1}} > 0$

After $n \geq 2$, the sequence becomes monotonically decreasing

Since the sequence is eventually monotonic and bounded below by 0,

By the Monotone Convergence Theorem, the sequence converges.

(2),First four terms:$a_1 = 1, a_2 = 1.5, a_3 = 1.75, a_4 = 1.875$

For $n=1, a_1 = 1 \leq 2$,

Assume the statement is true for $n=k, a_k \leq 2$

$a_{k+1} = 1 + \frac{1}{2}a_k \leq 1 + \frac{1}{2} \times 2 = 2$

By mathematical induction, $a_n \leqslant 2$ for all n,

Since $a_{n+1} - a_n = 1 - \frac{1}{2}a_n \geq 0$ (since $a_n \leq 2$)

The sequence is monotonically increasing,

By the Monotone Convergence Theorem, the sequence converges.

18. Find the convergence set of the power series $\Sigma_{n=1}^{\infty} \frac{(-1)^{n-1}}{n \cdot 3^n} x^n$

【Solution】

Let $a_n = \frac{(-1)^{n-1}}{n \cdot 3^n}$,then $\rho = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{1}{3} \frac{n}{n+1} = \frac{1}{3}$

The radius of convergence is $R = \frac{1}{\rho} = 3$, so the initial interval is (-3, 3)

Check endpoints:

At x = 3, the series becomes $\Sigma_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$, an alternating harmonic series, which converges by the Leibniz test.

At x = -3, the series becomes $\Sigma_{n=1}^{\infty} \frac{-1}{n}$, a harmonic series, which diverges.

Thus, the convergence set is (-3,3]

19. Find the sum function S(x) of the power series $\sum_{n=1}^{\infty} nx^{n-1}$ and determine its convergence set.

【Solution】

$\rho = \lim_{n\to\infty} \frac{n+1}{n} = 1$,so R = 1

At $x = \pm 1$, the general term does not approach 0, so the series diverges. The convergence set is (-1,1)

Because |x| < 1, so $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$

Differentiating term-by-term:

$$\frac{d}{dX}\sum_{n=0}^{\infty} x^n = \sum_{n=1}^{\infty} nX^{n-1} = \frac{1}{(1-x)^2}$$

Thus, the sum function is S(x) = $\frac{1}{(1-x)^2}$ with domain $x \in (-1,1)$

20. Expand the function $f(x) = \frac{1}{(1-x)^2}$ as a power series in x and find its convergence set.

【Solution】

Because geometric series $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n (|x| < 1)$

Differentiating term-by-term:

$$\frac{d}{dx}\left(\frac{1}{1-x}\right) = \frac{1}{(1-x)^2} = \Sigma_{n=1}^{\infty} nx^{n-1}$$

Let k = n – 1,so $\Sigma_{k=0}^{\infty}(k+1)x^k$

Determine the convergence set:

The original geometric series converges on (-1,1) and the endpoints $x \pm 1$,diverge since the general term does not approach 0.

Thus, the expansion is $\frac{1}{(1-x)^2} = \Sigma_{n=0}^{\infty}(n+1)x^n$ with convergence set (-1,1)

21. Find the limit of the function $\lim_{x\to 0} \frac{x^2 - x\ln(1+x)}{\sqrt{1+\tan x} - \sqrt{1+\sin x}}$

【Solution】

By using the Taylor series,

$$= \lim_{x\to 0} \frac{x^2 - x\left(x - \frac{x^2}{2} + o(x^2)\right)}{(1+\tan x)^{\frac{1}{2}} - (1+\sin x)^{\frac{1}{2}}} = \lim_{x\to 0} \frac{\frac{x^3}{2}}{1 + \frac{1}{2}\tan x - 1 - \frac{1}{2}\sin x}$$

$$= \lim_{x\to 0} \frac{\frac{x^3}{2}}{\frac{1}{2}\tan x - \frac{1}{2}\sin x} = \lim_{x\to 0} \frac{x^3}{\tan x(1-\cos x)} = \lim_{x\to 0} \frac{x^3}{x \cdot \frac{1}{2}x^2} = 2.$$

22. Find the convergence domain of the series,

$$\sum_{n=1}^{\infty}\left[1-n\ln\left(1+\frac{1}{n}\right)\right]x^n$$

【Solution】

To find the convergence interval,

$$\lim_{n\to\infty}\frac{\left|1-(n+1)\ln\left(1+\frac{1}{n+1}\right)\right|}{\left|1-n\ln\left(1+\frac{1}{n}\right)\right|}=\frac{1}{R}$$

Suppose,

$$a_n=1-n\ln\left(1+\frac{1}{n}\right)$$

By using Taylor series,

$$1-n\ln\left(1+\frac{1}{n}\right)=1-n\left(\frac{1}{n}-\frac{1}{2n^2}+o\left(\frac{1}{n^2}\right)\right)=\frac{1}{2n}+o\left(\frac{1}{n}\right)$$

So,

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim_{n\to\infty}\left|\frac{\frac{1}{2n+2}+o\left(\frac{1}{n}\right)}{\frac{1}{2n}+o\left(\frac{1}{n}\right)}\right|=1=\frac{1}{R}$$

The convergence interval is (-1,1) .

When x=1, the series is positive series

$$\sum_{n=1}^{\infty}\left[1-n\ln\left(1+\frac{1}{n}\right)\right]$$

Since,

$$\lim_{n\to\infty}\frac{1-n\ln\left(1+\frac{1}{n}\right)}{\frac{1}{2n}}=1$$

The positive series has the same divergence to $\sum_{n=1}^{\infty}\frac{1}{2n}$

When x= -1,the series is alternating series:

$$\sum_{n=1}^{\infty}(-1)^n\cdot\left[1-n\ln\left(1+\frac{1}{n}\right)\right]$$

Then we can find the monotonicity of $1-nln(1+\frac{1}{n})$

Let,

$$F(x)=1-x\ln\left(1+\frac{1}{x}\right)=1-x\ln(1+x)+x\ln x$$

$$F'(x)=-\ln(1+x)-\frac{x}{1+x}+\ln x+1=\frac{1}{1+x}-\ln\left(1+\frac{1}{x}\right)$$

$$F''(x) = \frac{1}{x(1+x)} - \frac{1}{(1+x)^2}$$

When x>0, F''(x)>0, F'(x) monotonic increase.
So,

$$F'(x) < \lim_{x \to +\infty} F'(x) = 0$$

When x>0, F(x) monotonic decrease.

Since

$$\lim_{n \to \infty} 1 - n\ln\left(1 + \frac{1}{n}\right) = 0$$

The alternating series converges.

The convergence domain is [-1,1) .

Part 2 往届真题解析

MTH004 Final Exams

【2021-2022 Final】

21. (1) Let z=ln(x−y), $x = se^t, y = e^{st}$. Find $\frac{\partial z}{\partial t}$ by the chain rule.

(2) Let z=z(x,y) be an implicit function defined by the equation $x^3 + 3y^2z - xyz^3 = 0$. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

22. Consider the Cobb-Donglas production model for a manufacturing process: P(x,y)$= kx^{\frac{1}{3}}y^{\frac{2}{3}}, (k > 0)$, where x and y are two inputs with unit costs 3 and 2 respectively. If the cost constraint is 3x+2y=54, determine x and y to maximize the production.

23. Find the volume of the solid bounded by z$= 2\sqrt{x^2 + y^2}$ and $z = 3 - (x^2 + y^2)$.

24. Let $I_n = \iint_S \frac{(\arcsin y)^{n-1}}{\sqrt{(1-x^2)(1-y^2)}} dA (n = 1,2,\cdots)$,where S is the plane region bounded by x−axis, y$=$ x and $x = \frac{1}{2}$.

(1) Find I_n.

(2) Let $a_n = (n + 1)I_n$. Find the convergence set and the sum of the power series $\sum_{n=1}^{\infty} a_n x^n$.

【2022-2023 Final】

19. Given the surface z=f(x,y)=$\sqrt{2+x^2-2y^2}$, and the point Q(1,1).

(1) Find the partial derivatives $\left.\frac{\partial z}{\partial x}\right|_{(1,1)}$ and $\left.\frac{\partial z}{\partial y}\right|_{(1,1)}$

(2) Let C be the curve intersecting the surface z=f(x,y) and the plane x=1. Find the parametric equations of the line tangent to C at P(1,1,1).

(3) Find the directional derivative of f(x,y) at Q in the direction u=2i−j

(4) In what direction does f increase most rapidly at Q ? What rate of change of f at the point Q in that direction?

20. Assuming that the marketing cost of online advertising is x (in million of yuan) and the marketing cost of television is y (in million of yuan), the company's revenue R is a function of x and y:

$$R(x,y)=20+10x+6y-xy-2x^2-y^2$$

(1) Find the local extreme value of R(x,y).

(2) If the budget for the total marketing costs is 2 (in million of yuan), how should this amount be allocated in order to maximize the revenue R?

21. (1) Evaluate the double integral $\iint_S \sqrt{x^2+xy}\,dA$, where S is the region bounded by the lines y=x, x=1, x=2 and y=0.

(2) Find the volume of the solid bounded by the surfaces z=12 $-2x^2-y^2$ and $z=x^2+2y^2$.

22. Consider the power series $\sum_{n=0}^{\infty}(n+2)x^n$

(1) Find the convergence set for this power series.

(2) Find the sum of this power series.

【2023-2024 Final】

19. Let f(x,y)=xe^{y^2}

(1) Find the gradient ∇f.

(2) Determine in which direction f INCREASES most quickly at (1,1). And at what rate?

(3) Using total differential to estimate f(1.1, 1.1).

(4) Find $\left[\frac{d}{du}f(\sqrt{u}, u^2)\right]_{u=1}$

20. Find the maximum and minimum value of f(x,y)= e^{xy} on D=$\{(x,y)\,|\,x^2+4y^2 \le 2\}$

21. (1)Evaluate the volume of the soild bounded by

$S_1: z = 20 - 4x^2 - y^2$ and $S_2: z = x^2 + 4y^2$

(2) Given Π: 8x+4y+z=28, find the area of Π in the first octant.

22. Find g(x) such that g(x) $= \sum_{n=1}^{\infty} \frac{(-1)^{n-1}x^{2n+1}}{2n(2n-1)}$

【2021-2022 Final】

21.

【Solution】

(1) $\frac{\partial z}{\partial t}=\frac{\partial z}{\partial x}\times\frac{\partial x}{\partial t}+\frac{\partial z}{\partial y}\times\frac{\partial y}{\partial t}$

(2) F(x,y,z)= $x^3+3y^2z-xyz^3=0$

$$\frac{\partial z}{\partial x}=-\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}}$$

22.

【Solution】

$\nabla f(x,y)=\lambda\nabla g(x,y)$ $\quad$ g(x,y)= $3x+2y-54=0$

$\langle ky^{\frac{2}{3}}\times\frac{1}{3}\times x^{-\frac{2}{3}},kx^{\frac{1}{3}}\times\frac{2}{3}\times y^{-\frac{1}{3}}\rangle=\lambda\langle 3,2\rangle$

23.

【Solution】

$2\sqrt{x^2+y^2}=3-(x^2+y^2)$

$x^2+y^2=1$ or $x^2+y^2=9$

$\quad\quad\quad

$$\lim_{n\to\infty}\frac{a_{n+1}}{a_n}=\lim_{n\to\infty}\frac{\frac{(\frac{\pi}{6})^{n+2}}{n+1}}{\frac{(\frac{\pi}{6})^{n+1}}{n}}=\lim_{n\to\infty}\frac{n}{n+1}\times\frac{\pi}{6}=\frac{\pi}{6}\qquad |x|<\frac{6}{\pi}$$

x= $\frac{6}{\pi}$: $\sum_{n=1}^{\infty}\frac{\frac{\pi}{6}}{n}$, $\lim_{n\to\infty}\frac{\frac{\frac{\pi}{6}}{n}}{\frac{1}{n}}=\frac{\pi}{6}$ $\sum_{n=1}^{\infty}\frac{1}{n}$ diverges, so $\sum_{n=1}^{\infty}\frac{\frac{\pi}{6}}{n}$ diverges.

x=$-\frac{6}{\pi}$: $\sum_{n=1}^{\infty}(-1)^n$, $\frac{\frac{\pi}{6}}{n}$ $\frac{\frac{\pi}{6}}{n}>\frac{\frac{\pi}{6}}{n+1}$ $\lim_{n\to\infty}\frac{\frac{\pi}{6}}{n}=0$, so $\sum_{n=1}^{\infty}(-1)^n\frac{\frac{\pi}{6}}{n}$ converges.

We can find the convergence set.

$$\sum_{n=1}^{\infty}(\frac{\pi}{6})^{n+1}\frac{x^n}{n}=\frac{\pi}{6}\sum_{n=1}^{\infty}\frac{1}{n}(\frac{\pi x}{6})^n$$

$$\ln(1+x)=x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}\cdots\cdots$$

$$\ln(1-x)=-x-\frac{x^2}{2}-\frac{x^3}{3}-\frac{x^4}{4}\cdots\cdots$$

$$-\ln(1-x)=x+\frac{x^2}{2}+\frac{x^3}{3}+\frac{x^4}{4}\cdots\cdots$$

$$\therefore -\ln(1-x)=\sum_{n=1}^{\infty}\frac{x^n}{n},$$

Let $x=\frac{\pi x}{6}$,We can find $\sum_{n=1}^{\infty}a_nx^n$.

【2022-2023 Final】

19.

【Solution】

(1)$\frac{\partial z}{\partial x}$ =2x $\times\frac{1}{2}\times\frac{1}{\sqrt{2+x^2-2y^2}}$ =$\frac{x}{\sqrt{2+x^2-2y^2}}$

$\frac{\partial z}{\partial y}$ =$-4y\times\frac{1}{2}\times\frac{1}{\sqrt{2+x^2-2y^2}}$ =$\frac{-2y}{\sqrt{2+x^2-2y^2}}$

将 x=1 y=1 代入即可

(2)已经限定在 x=1 的平面内，所以切线方程不含 x

z=f(1,y)=$\sqrt{3-2y^2}$

z'=$\frac{-2y}{\sqrt{3-2y^2}}$ y=1,z=-2

直线方向向量$(0,1,-2)$ 直线过点 P 即可求出切线方程

(3)$\nabla f=\langle f_x,f_y\rangle=\langle\frac{x}{\sqrt{2+x^2-2y^2}},\frac{-2y}{\sqrt{2+x^2-2y^2}}\rangle$

将 u 化为单位向量：a=$\langle\frac{2}{\sqrt{5}},-\frac{1}{\sqrt{5}}\rangle$

$\nabla f(1,1)\cdot a$

(4)f(x, y)在梯度方向增长最快

将$\nabla f_{(1,1)}$单位化即为所求

rate：将$\nabla f_{(1,1)}$与刚求出的单位向量点乘，也就是$\|\nabla f\|$

20.

【Solution】

(1) $R_x = 10 - y - 4x \quad R_y = 6 - x - 2y$

$$\begin{cases} 10 - y - 4x = 0 \\ 6 - x - 2y = 0 \end{cases} \Rightarrow \begin{cases} x = 2 \\ y = 2 \end{cases}$$

$R_{xx} = -4 \;\; R_{yy} = -2 \;\; R_{xy} = -1$

D=7> 0

Let x=2 y=2

(2) $\nabla f(x,y) = \lambda \nabla g(x,y) \quad g(x,y) = x + y - 2 = 0$

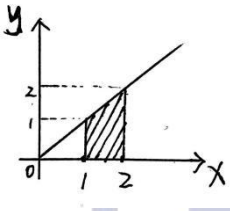
$\langle 10 - y - 4x, 6 - x - 2y \rangle = \lambda \langle 1,1 \rangle$

21.

【Solution】

(1)

$\iint_S \sqrt{x^2 + xy}\, dA$



$\sqrt{x^2 + xy}$ y 出现的比 x 少 所以先积 y

对于每一个 x，y 从 0 积到 x，$1 \le x \le 2$

$\int_1^2 \int_0^x \sqrt{x^2 + xy}\, dydx$

$= \int_1^2 \left[\frac{2}{3x}(x^2 + xy)^{\frac{3}{2}}\right]_0^x dx$

$= \int_1^2 \frac{2(2x^2)^{\frac{3}{2}}}{3x} - \frac{2x^3}{3x} dx$

$= \int_1^2 \frac{2x^2}{3}(2^{\frac{3}{2}} - 1) dx$

$= \left[\frac{2x^3}{9}(2^{\frac{3}{2}} - 1)\right]_1^2$

(2)$12 - 2x^2 - y^2 = x^2 + 2y^2$

$x^2 + y^2 = 4$

$$\iint_S 12 - 2x^2 - y^2 - (x^2 + 2y^2)dA$$

$$= \iint_S 12 - (3x^2 + 3y^2)dA$$

$$= \int_0^{2\pi}\int_0^2 (12 - 3r^2)\, rdrd\theta$$

$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}$$

r=2

22.

【Solution】

(1)

$$\lim_{n\to\infty}\frac{n+3}{n+2} = 1 \qquad |x| < 1$$

x=1

$$\sum_{n=0}^{\infty} n + 2 > \sum_{n=0}^{\infty} n$$

$\sum_{n=0}^{\infty} n$ diverges, so $\sum_{n=0}^{\infty} n + 2$ diverges

x=−1

$$\sum_{n=0}^{\infty}(n+2)(-1)^n$$

$a_n = n + 2$ $a_{n+1} > a_n$ $\lim_{n\to\infty} a_n = \infty$ diverges

(2)

$$\sum_{n=0}^{\infty}(n+2)x^n = \sum_{n=0}^{\infty}(n+1)x^n + \sum_{n=0}^{\infty}x^n$$

$$\int_0^x \sum_{n=0}^{\infty}(n+1)x^n\, dx = \sum_{n=0}^{\infty}\int_0^x (n+1)x^n dx = \sum_{n=0}^{\infty} x^{n+1} = \frac{1}{1-x} - 1$$

$$\sum_{n=0}^{\infty}(n+1)x^n = \left(\frac{1}{1-x} - 1\right)' = \frac{1}{(1-x)^2}$$

【2023-2024 Final】

19.

【Solution】

(1) $f_x = e^{y^2}$ $f_y = 2xye^{y^2}$

$\nabla f = \langle f_x, f_y \rangle$

(2) $\nabla f_{(1,1)} = \langle e, 2e \rangle$

$v = \frac{\nabla f_{(1,1)}}{\|\nabla f_{(1,1)}\|}$

$v \cdot \nabla f_{(1,1)} = \|\nabla f_{(1,1)}\|$

f(x, y)在梯度方向增长最快

将$\nabla f_{(1,1)}$单位化即为所求

rate：将$\nabla f_{(1,1)}$与刚求出的单位向量点乘，也就是$\|\nabla f\|$

(3) $z=xe^{y^2}$

$dz=f_x dx+f_y dy$

dz=f(1.1, 1,1)−f(1,1) dx=1.1−1 dy=1.1−1

$f_x = e^{y^2}=e$ $f_y = 2xye^{y^2}=2e$

f(1.1, 1,1)−f(1,1)=e× $0.1 - 2e \times 0.1$=0.3e

(4) $\frac{d}{du} f(\sqrt{u}, u^2) = (\sqrt{u})' \times f_{\sqrt{u}} + (u^2)' f_{u^2}$

复合函数求导，先对两个变量求导，再求偏导，求偏导就是把原来f_x f_y 中的 x y 分别换成$\sqrt{u}$ u^2 最后把 x=1 代入

20.

【Solution】

boundary

$\nabla f(x,y) = \lambda \nabla g(x,y)$ $g(x,y) = x^2 + 4y^2 - 2 = 0$

$\langle ye^{xy}, xe^{xy} \rangle = \lambda \langle 2x, 8y \rangle$

→ $(1, \frac{1}{2})$ $(-1, -\frac{1}{2})$

internal

$\begin{cases} f_x = ye^{xy} = 0 \\ f_y = xe^{xy} = 0 \end{cases}$ → x=y=0 → (0,0)

把五个点代入 f(x, y)=e^{xy}通过比较函数值来确定最大最小值

21.

【Solution】

(1) $20-4x^2-y^2=x^2+4y^2$

$$x^2+y^2=4$$

$$V=\iint[(20-4x^2-y^2)-(x^2+4y^2)]dA$$

$$=\iint(20-5x^2-5y^2)dA$$

$$=\int_0^{2\pi}\int_0^2(20-5r^2)rdrd\theta$$

$$\begin{cases}x=r\cos\theta\\ y=r\sin\theta\end{cases}$$

r=2

（2） Let x=y=0, The plane intersects with the z-axis at A (0,0,28)

Similarly, the plane intersects with the x-axis and y-axis at B(4,0,0) C(0,7,0), respectively.

$\overrightarrow{AB}=\langle 4,0,-28\rangle$ $\overrightarrow{AC}=\langle 0,7,-28\rangle$

$S=\frac{1}{2}\times\|\overrightarrow{AB}\|\times\|\overrightarrow{AC}\|\times\sin\langle\overrightarrow{AB},\overrightarrow{AC}\rangle=\frac{1}{2}\|\overrightarrow{AB}\times\overrightarrow{AC}\|$

22.

【Solution】

$$\sum_{n=1}^{\infty}(-1)^{n-1}\left[\frac{x^{2n+1}}{2n-1}-\frac{x^{2n+1}}{2n}\right]$$

$$=\sum_{n=1}^{\infty}(-1)^{n-1}\frac{x^{2n+1}}{2n-1}-\sum_{n=1}^{\infty}(-1)^{n-1}\frac{x^{2n+1}}{2n}$$

$$=x^3-\frac{x^5}{3}+\frac{x^7}{5}\cdots\cdots-(\frac{x^3}{2}-\frac{x^5}{4}+\frac{x^7}{6}\cdots\cdots)$$

$$=x^2(x-\frac{x^3}{3}+\frac{x^5}{5}\cdots\cdots)-\frac{x}{2}(x^2-\frac{x^4}{2}+\frac{x^6}{3}\cdots\cdots)$$

$$=x^2(x-\frac{x^3}{3}+\frac{x^5}{5}\cdots\cdots)-\frac{x}{2}((x^2)^1-\frac{(x^2)^2}{2}+\frac{(x^2)^3}{3}\cdots\cdots)$$

MTH004 Resit Exams

【2021-2022 Final】

19.(12 marks)

(1) Let z=u^2lnv ,u=$\frac{x}{y}$,v=3x-2y.Find $\frac{\partial z}{\partial y}$ by the chain rule.

(2) Let z=z(x,y) be an implict function defined by the equation e^z-xyz=0 .Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

20.(14 marks)

(1)Find the local maximum and minimum values, if any, of f(x,y)=x^3+y^3+3xy.

(2)Let Ω be a rectangular box without a lid.Find the dimensions of Ω with surface area 108ft^2 that has maximum volume and find the maximum volume.

21.(12 marks)

(1)Find the double integral $\iint_s xydA$, where s is the region bounded by x^2+y^2=2y, y=x and y-axis.

(2)Find the volume of the solid in the first octant bounded by the coordinate planes and the planes 2x+y-4=0 and 8x+y-4z=0.

22.(8 marks)

Let S be the region bounded by y=x,x²+y²=x and x-axis.

$$a_n = \iint_s \frac{y^n}{(x^2+y^2)^{\frac{n+1}{2}}} dA$$

Find $\sum_{n=0}^{\infty} a_n$.

【2022-2023 Final】

19.(12 marks) Let f(x,y,z)=$x^3e^{y^2z}$,P(-1,1,1).

(1)Find the gradient of f(x,y,z) at P.

(2)Find the directional derivative of f(x,y,z)at P in the direction u=2i-j+2k.

(3)In what direction does f increase most rapidly at P? What is the rate of change of f at the point P in that direction?

(4)In what direction the directional derivative of f(x,y,z) at P is zero?

20.(12 marks) Find the global maximum value and global minimum value of

z= f(x,y) =x^2+3y^2-4x on the set S={(x,y)|$(x-1)^2$+y^2≤9}.

21.(12 marks)

(1)Evaluate the double integral $\iint_S (x+y)dA$,where S is the region enclosed by the triangle with vertices (0,0),(1,4) and(2,4).

(2)Find the volume of the solid bounded by the surface z=$e^{x^2+y^2}(x^2+y^2)$, the plane z=0 and the cylinder x^2+y^2=1.

22.(10 marks)

(1)Find the Maclaurin series representation of f(x)=ln(2+3x-2x^2).

(2)Find the convergence radius and convergence set for this Maclaurin series.

【2023-2024 Final】

19. (12 marks) Consider upper half of the two-sheet hyperboloid S described by function z = f (x, y) =$\sqrt{x^2+3y^2+3}$ and a point P (1,2,4) on the surface.

(1)Calculate $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at P.

(2)Find the equation of tangent plane Π of S at point P.

(3)Find the parametric equations of the intersection line of Π and the plane y=2.

20.(12 marks)

(1) Given three non-zero numbers a, b, c. Show that the equation of the plane which intersect coordinate axis with (a,0,0) ,(0, b,0) ,(0,0, c) is $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$.

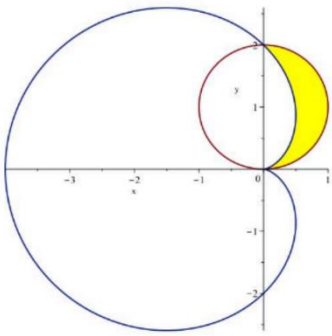
(2) Assume that a>0, b>0 and c>0. Find the values of a, b, c such that the plane passes through the point (3,2,1) and cuts off the smallest volume in the first octant. (Hint: Volume of a tetrahedron can be computed by V= $\frac{1}{3}$ (area of base)·(height).)

21.(12 marks) Let $\gamma 1$: $x^2+(y-1)^2=1$ and $\gamma 2$: $x^2+y^2-2\sqrt{x^2+y^2}+2x=0$ be curves on the xy-plane. Let D be the region that outside $\gamma 2$ but inside $\gamma 1$.

(1) Rewrite $\gamma 1$ and $\gamma 2$ in polar coordinates.

(2) Find the area of D (colored in yellow below).

(3) Find the volume between f(x,y)= $\begin{cases} \frac{x+y}{x^2+y^2}, (x,y) \neq (0,0) \\ 0, (x,y) = (0,0) \end{cases}$ and the xy-plane over D.



22.(10 marks) Does $\lim_{x\to 0} \frac{\tan x^{-1}(x^3)+x\cos x-\sum_{n=1}^{\infty}\frac{x^{2n-1}}{n}}{x^5}$ exist? Calculate the limit if so or state why the limit does not exist.

【2021-2022 Final】

19.

(1) 【辅助理解】使用链式法则，将z表示为u和v的函数，再分别求u和v对y的偏导

【Solution】

$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\frac{\partial u}{\partial y}+\frac{\partial z}{\partial v}\frac{\partial v}{\partial y}=2u\ln v(-\frac{x}{y^2})+\frac{u^2}{v}(-2)$

(2) 【辅助理解】本题考察隐函数求导

【Solution】

$e^z\frac{\partial z}{\partial x}-yz-xy\frac{\partial z}{\partial x}=0\rightarrow\frac{\partial z}{\partial x}=$

$e^z\frac{\partial z}{\partial y}-xz-xy\frac{\partial z}{\partial y}=0\rightarrow\frac{\partial z}{\partial y}=$

20.

(1) 【辅助理解】本题 考察寻找局部极值：求梯度并解临界点，二阶偏导判断极值类型

【Solution】

$f_x=3x^2+3y$

$f_y=3y^2+3x$

Let $f_x=0$，$f_y=0$，（0，0），（-1，-1）

$f_{xx}=6x$，$f_{xy}=3$，$f_{yy}=6x$

D（0，0）$=f_{xx}f_{yy}-f_{xy}^2=$

f_{xx}（0，0）=

D（-1，-1）$=f_{xx}f_{yy}-f_{xy}^2=$

f_{xx}（-1，-1）=

Then compare the values f_{xx} and 0 ，D（x，y） and 0 ，to determine the local maximum and minimum values

(2) 【辅助理解】本题考察拉格朗日乘数法：设长方体长宽高为x，y，z，表面积约束xy+2yz+2xz=108，求V max=xyz

【Solution】

Let F=xyz+λ（108-xy-2xz-2yz）

Find $\frac{\partial F}{\partial x}$,$\frac{\partial F}{\partial y}$,$\frac{\partial F}{\partial z}$

Let $\frac{\partial F}{\partial x}$,$\frac{\partial F}{\partial y}$,$\frac{\partial F}{\partial z}$=0

Get x，y，z，then V max=xyz

21.

(1)【辅助理解】本题考查二重积分在极坐标系下的计算

积分区域 s 由圆$x^2+y^2=2y$，直线 y=x 和 y 轴围成，将圆方程化为极坐标形式：

$r=2\sin\theta$，y=x 对应$\theta=\frac{\pi}{4}$，积分区域为$\theta\in[0,\frac{\pi}{4}]$，$r\in[0,2\sin\theta]$。在极坐标下，被积函数 $xy=r^2\cos\theta\sin\theta$，$dA=rdrd\theta$，然后计算积分。

【Solution】

$$\iint_s xydA=\int_0^{\frac{\pi}{4}}\int_0^{2\sin\theta} r^3\cos\theta\sin\theta\, drd\theta =\int_0^{\frac{\pi}{4}} 4\sin^4\theta\cos\theta\sin\theta=$$

(2)【辅助理解】本题考查二重积分，先确定积分区域：平面 2x+y-4=0在xy平面的投影为三角形区域，$0\le x\le 2$，$0\le y\le 4-2x$，高度函数：$z=2x+\frac{y}{4}$。

【Solution】

$$V=\iint_D \left(2x+\frac{y}{4}\right) dA=\int_0^2\int_0^{4-2x}\left(2x+\frac{y}{4}\right)dydx =\int_0^2\left(8x-4x^2+\frac{(4-2x)^2}{8}\right)dx=$$

22.

【辅助理解】先确定积分区域：S 由 $y=x, x^2+y^2=x$ 和 x 轴围成，将圆方程转化为极坐标→$r=\cos\theta$ （$0\le\theta\le\frac{\pi}{4}$），积分区域为 $0\le\theta\le\frac{\pi}{4}$，$0\le r\le\cos\theta$。

在极坐标下，简化被积函数，计算积分，再进行级数求和。

【Solution】

$$a_n=\iint_s \frac{y^n}{(x^2+y^2)^{\frac{n+1}{2}}}\, dA=\iint_s \frac{(r\sin\theta)^n}{(r)^{n+1}} r\, drd\theta=\int_0^{\frac{\pi}{4}}\int_0^{\cos\theta}\sin^n\theta\, dr\, d\theta=\int_0^{\frac{\pi}{4}}\cos\theta\sin^n\theta\, d\theta$$

Let $u=\sin\theta$，$du=\cos\theta d\theta$，$a_n=\int_0^{\frac{\sqrt{2}}{2}} u^n du=\frac{1}{n+1}\left(\frac{\sqrt{2}}{2}\right)^{n+1}$

$$\sum_{n=0}^{\infty} a_n=\sum_{n=0}^{\infty}\frac{1}{n+1}\left(\frac{\sqrt{2}}{2}\right)^{n+1}=$$

【2022-2023 Final】

19.

(1)【辅助理解】本题考查梯度计算（12.5），$\nabla f=\left(\frac{\partial f}{\partial x},\frac{\partial f}{\partial y},\frac{\partial f}{\partial z}\right)$

【Solution】

$$\frac{\partial f}{\partial x}=3x^2e^{y^2z}$$

$$\frac{\partial f}{\partial y}=x^3\cdot 2yze^{y^2z}$$

$$\frac{\partial f}{\partial z}=x^3\cdot y^2e^{y^2z}$$

Substitute P(-1,1,1)

(2)【辅助理解】本题考查方向导数（12.5），$D_uf=\nabla f\cdot\frac{u}{\|u\|}$

【Solution】

$$\|u\|=\sqrt{2^2+(-1)^2+2^2}=3$$

$\frac{u}{\|u\|}=(\frac{2}{3},-\frac{1}{3},\frac{2}{3})$

$D_u f=\nabla f\cdot\frac{u}{\|u\|}$

(3)【辅助理解】(12.5)函数在某点最快上升方向是其梯度方向，变化率是梯度的模长

【Solution】

From (1),$\nabla f=(\frac{\partial f}{\partial x},\frac{\partial f}{\partial y},\frac{\partial f}{\partial z})$

$\|\nabla f\|=\sqrt{\frac{\partial f}{\partial x}^2+\frac{\partial f}{\partial y}^2+\frac{\partial f}{\partial z}^2}$

(4)【辅助理解】(12.5)方向导数为零的条件是方向向量与梯度正交，找出满足3a-2b-c=0的非零向量即可

【Solution】

Let v=(a,b,c)

$\nabla f\cdot v=0$

20.

【辅助理解】

本题求全局最大值和最小值，需同时检查函数在区域内部和边界的极值。步骤包括：

1. 寻找内部临界点[驻点：梯度为零的点，奇异点：函数不可导或不连续的点（但本题函数为多项式函数，显然不存在奇异点，故不做讨论）]

2. 检查边界点（拉格朗日乘数法）

3. 比较所有值，确定全局最大值和最小值

【Solution】

$f(x,y)=x^2+3y^2-4x$

$\nabla f=(\frac{\partial f}{\partial x},\frac{\partial f}{\partial y})=(2x-4, 6y)$

Let $\nabla f=0,\begin{cases}2x-4=0\\6y=0\end{cases},\rightarrow x=2,y=0$

The stationary point is(2,0),f(2,0)= -4

$g(x,y)=(x-1)^2+y^2-9=0$

$\nabla g=(2(x-2),2y)$

From $\nabla f = \lambda \nabla g$, $\begin{cases} 2x-4=\lambda\cdot 2(x-1) & ① \\ 6y=\lambda\cdot 2y & ② \\ (x-1)^2+y^2=9 & ③ \end{cases}$

From ②,y(6-2λ)=0

(1) y=0,substitute y=0 into ③
solving the equation, we obtain x=4 or x=-2
f(4,0)=0,f(-2,0)=12

(2) λ=3,substitute λ=3 into ①
2x-4=6(x-1)
x=0.5

substitute x=0.5 into ③,we obtain y=$\pm\frac{\sqrt{35}}{2}$

f(0.5,$\pm\frac{\sqrt{35}}{2}$)=24.5

Then Compare all values to determine the global maximum and minimum values.

21.

(1) 【辅助理解】将三角形划分为两个子区域，分别确定积分限→在 x∈[0, 1]和 x∈[1, 2]内对 y 分段积分

【Solution】

For sub-region 1: $\int_0^1\int_{2x}^{4x} xydydx=\int_0^1 6x^3dx$

For sub-region 2:$\int_1^2\int_{2x}^{4} xydydx=\int_1^2 (8x-2x^3)dx$

Sum up

(2) 【辅助理解】本题求由曲面z=$e^{x^2+y^2}(x^2+y^2)$、平面z=0和圆柱$x^2+y^2=1$所围成的立体的体积。首先，因为这里z的表达式已经给出，且底面是z=0，所以更适合使用二重积分，可在xy平面上的投影区域上进行积分，积分函数就是z的高度。

其次，圆柱 $x^2+y^2=1$在xy平面上的投影是半径为1的圆，因此积分区域D为$x^2+y^2\leqslant 1$，z=0则是底面，上面的曲面是z=$e^{x^2+y^2}(x^2+y^2)$。所以体积V是这个曲面在区域D上的积分，即V=$\iint De^{x^2+y^2}(x^2+y^2)dA$。

考虑到积分区域D是圆形的，所以使用极坐标会比较方便。在极坐标下，$x^2+y^2=r^2$，dA=r dr dθ。积分区域D变为r∈[0, 1]， θ∈[0, 2π]。

【Solution】

The region of integration is the unit disk D:0≤r≤1，0≤θ≤2π

$V=\iint_D e^{r^2}r^2\ dA=\int_0^{2\pi}\int_0^1 e^{r^2}r^2\ r\,dr\,d\theta$

$V=2\pi\int_0^1 e^{r^2}r^3 dr\,d\theta$

Let $t=r^2$,so $dt=2rdr$:$V=\pi\int_0^1 te^t dt$

Using integration by parts with u=t, $dv=e^t dt$: $\int te^t dt=te^t-e^t+C$

Evaluating it from 0 to 1,then get the volume.

22.

(1) 【辅助理解】本题求解麦克劳林级数展开式（9.8）

【Solution】We know that the Maclaurin series expansion of ln(1+x) is:

$$x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\ldots$$

We can rewrite f(x) as:

$$f(x)=\ln[2(1+\frac{3}{2}x-x^2)]$$

$$f(x)=\ln2+[\frac{3}{2}x-x^2-\frac{(\frac{3}{2}x-x^2)^2}{2}+\frac{(\frac{3}{2}x-x^2)^3}{3}-\cdots]$$

$$f(x)=\ln2+\frac{3}{2}x-x^2-\frac{9}{8}x^2+\cdots$$

$$f(x)=\ln2+\frac{3}{2}x-\frac{17}{8}x^2+\cdots$$

So, the Maclaurin series representation of f(x) is:

(2) 【辅助理解】求该级数的收敛半径和收敛集（9.6），步骤为：1，确定收敛半径；2，检查端点；3，确定收敛集。

【Solution】$R=\lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right|$

Where a_n is the coefficient of the x^n term in the series.

$a_0=\ln2$

$a_1=\frac{3}{2}$

$a_2=-\frac{17}{8}$

Applying the ratio test: $R=\lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right|=\lim_{n\to\infty}\left|\frac{a_n}{a_n}\right|=1$

So, the convergence radius R=1

The convergence set for this Maclaurin series is the interval of convergence, which extends from the center of the series,x=0, to a distance of R in either direction.

Thus, the convergence set is-1<x<1.

【2023-2024 Final】

19.

（1）【辅助理解】本题考察计算偏导数（12.2），使用链式法则对z = f (x, y) $=\sqrt{x^2+3y^2+3}$ 分别求 $\frac{\partial z}{\partial x}$和 $\frac{\partial z}{\partial y}$：

【Solution】

$$\frac{\partial z}{\partial x} = \frac{x}{\sqrt{x^2+3y^2+3}}$$

$$\frac{\partial z}{\partial y} = \frac{3y}{\sqrt{x^2+3y^2+3}}$$

Substitute P(1,2,4):

（2）【辅助理解】切平面方程公式（12.7）为$z-z_0=f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$，代入偏导数值和点P(1,2,4)

【Solution】

The equation of the tangent plane is:$z-z_0=f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$

Substitute P(1,2,4) and $f_x(1,2)$, $f_y(1,2)$

Simplify to standard form

（3）【辅助理解】将y=2代入切平面方程，解出z关于x的表达式

设 x=t，将 z 用含 t 的参数表示，可得出交线的参数方程

【Solution】

Substitute y=2 into the tangent plane equation

Let x=t, the parametric equations are:$\begin{cases} x = t \\ y = 2 \\ z = \end{cases}$,t ∈R

20.

(1) 【辅助理解】本题考查平面方程的推导（11.8），可利用截距点直接构造平面方程

【Solution】

The intercept form of the plane equation: $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

Substitute the point (a,0,0) ,(0, b,0) ,(0,0, c)

(2) 【辅助理解】平面的一般截距式是 x/a + y/b + z/c = 1，该平面过点(3, 2, 1)，所以代入后得 3/a+ 2/b+ 1/c= 1。

题目的目标是最小化该体积，V=(1/3)×(1/2)ab×c =(1/6)abc，即最小化 a×b×c，同时满足约束条件 3/a+ 2/b+ 1/c= 1。

构造拉格朗日函数 L = abc + λ(1 −3/a −2/b −1/c)。然后对 a、b、c 求偏导，并令其等于0。通过消去 λ 可得 a, b, c 的比例关系，带入约束条件即可求出 a, b, c，得出 V。

【Solution】

The intercept form of the plane in the first octant is:

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$

Since the plane passes through (3,2,1), substituting this point gives the constraint: $\frac{3}{a}+\frac{2}{b}+\frac{1}{c}=1$

The tetrahedron formed by the plane and the coordinate axes has volume:

V=$\frac{1}{6}$abc

Define the Lagrangian function:L(a, b, c, λ)= abc + λ(1−$\frac{3}{a}+\frac{2}{b}+\frac{1}{c}$)

Take partial derivatives and set them to zero: $\begin{cases} bc-\frac{3\lambda}{a^2}=0 \\ ac-\frac{3\lambda}{b^2}=0 \\ ab-\frac{3\lambda}{c^2}=0 \end{cases}$

a=3c,b=2c

Substitute a=3c,b=2c into the constraint: c=3

V=$\frac{1}{6}$abc=

21.

(1) 【辅助理解】γ_1：已知圆的方程 $x^2+(y-1)^2=1$，代入x=rcosθ,y=rsinθ，展开，γ_2同理

(2) 【辅助理解】由（1）得 γ_1和 γ_2的极坐标方程，联立解得θ_1和θ_2

Area= $\frac{1}{2}\int$（r 外^2 − r 内^2）dθ

【Solution】

2sinθ=2(1-cosθ)

Area = $\frac{1}{2}\int_{\theta_2}^{\theta_1}\{(2\sin\theta)^2-[2(1-\cos\theta)]^2\}d\theta$

(3) 【辅助理解】被积函数 f(x,y)= $\begin{cases}\frac{x+y}{x^2+y^2},(x,y)\neq(0,0)\\ 0,(x,y)=(0,0)\end{cases}$在极坐标下化简为: $\frac{\cos\theta+\sin\theta}{r}$

【Solution】

V=$\iint_D f(x,y)dA=\int_0^{\frac{\pi}{2}}\int_{2(1-\cos\theta)}^{2\sin\theta}\frac{\cos\theta+\sin\theta}{r}rdrd\theta$

$$=\int_0^{\frac{\pi}{2}}(\cos\theta+\sin\theta)[2\sin\theta-2(1-\cos\theta)]d\theta$$

=2$\int_0^{\frac{\pi}{2}}[(\cos\theta+\sin\theta)^2-(\cos\theta+\sin\theta)]d\theta$

=2$\int_0^{\frac{\pi}{2}}(1+\sin2\theta-\cos\theta-\sin\theta)d\theta$

22.

【辅助理解】本题涉及泰勒展开(9.8)和极限计算(12.3)

【Solution】

Expand Each Component in the Numerator via Taylor Series:

$\tan x^{-1}(x^3)$=x^3- $\frac{x^9}{3}$+⋯(higher-order terms neglected as x→0)

xcosx=x(1- $\frac{x^2}{2}$+ $\frac{x^4}{24}$ -⋯) =x- $\frac{x^3}{2}$+ $\frac{x^5}{24}$ -⋯

$\sum_{n=1}^{\infty}\frac{x^{2n-1}}{n}$ = x+$\frac{x^3}{2}$+$\frac{x^5}{3}$+⋯

Substitute the expansions into the numerator:

$\tan x^{-1}(x^3)+x\cos x-\sum_{n=1}^{\infty}\frac{x^{2n-1}}{n}$= x^3+x- $\frac{x^3}{2}$+ $\frac{x^5}{24}$ +x+$\frac{x^3}{2}$+$\frac{x^5}{3}$ =- $\frac{7x^5}{24}$

$\lim_{x\to0}\frac{\tan x^{-1}(x^3)+x\cos x-\sum_{n=1}^{\infty}\frac{x^{2n-1}}{n}}{x^5}$ = $\lim_{x\to0}\frac{-\frac{7x^5}{24}}{x^5}$ =

The limit exists and is equal to:

MTH008 Final Exams

【2021-2022 Final】

19.For $u = (2,1,6), v = (-2,1,0)$, $(u \cdot v)v =$ _____

20.Let $S = \sum_{n=0}^{\infty} \frac{3^n+2^n}{4^n}$,Then,S=_____

21. Evaluate $\int_{-1}^{1}\int_{0}^{3}\left(x^5y + xe^{xy}\right)dydx$=_____

22.Suppose $z = z(x,y)$ is implicitly determined by $\ln(x+y+z) = x + 2y + 3z$. Then $\frac{\partial_z}{\partial_y}|_{(x,y,z)=(-1,5,-3)}$=_____

23.Let $z = f(x,y) = x^y$,At $(x,y) = (3,2)$, if $dx = \frac{1}{10}, dy = -\frac{1}{5}$, then $dz =$ _____

24.Let $f(x,y) = 4x - 3y + 2$ and $S = \{(x,y): 2x^2 + 3y^2 \le 1\}$.Find the maximum and minimum value of f over the region S.

25.Let $f(x) = \frac{x+3}{3x^2-7x+2}$

(1)Find the partial fraction decomposition of f(x).
(2)Find the Taylor series of f(x) in x-1.Indicate the convergence set.

26.Let $B = \{(x,y,z): x^2 + y^2 + z^2 \le 1\}$ be solid sphere of radius 1, $u(x,y,z)$ be the distance from (x,y,z) to P(0,0,1).
(1) Find $u(x,y,z)$ and simplify it in the spherical coordinates:

$$x = \rho\sin\emptyset\cos\theta, y = \rho\sin\emptyset\sin\theta, z = \rho\cos\emptyset$$

(2)Convert $\iiint_B u(x,y,z)dV$ into an iterated integral in the spherical coordinates, in the order $d\emptyset d\rho d\theta$.

(3)Find the average distance m from B to P: $m = \frac{\iiint_B u(x,y,z)dV}{V_B}$, V_B=volume of B

【2022-2023 Final】

19.Consider a paraboloid with equation $z = x^2 + y^2$ and a sphere with equations $x^2 + y^2 + z^2 = 2$.

(1)Find the area of the part of the paraboloid , which is inside the sphere.

(2)Find the volume of the solid, which is bounded below by the paraboloid and above the sphere.

20.Consider a cylinder with equation $x^2 + y^2 = 1$ and a sphere with equation $(x-1)^2 + (y-1)^2 + (z-1)^2 = 1$.

(1)Find the tangent line equation of the intersection curve at point (0,1,1)

(2)Find the equation of plane tangent to both the sphere and the cylinder

21.Find the extreme values of $f(x,y) = x^4 + y^4 - 4xy + 1$ on the closed and bounded set $S = \{(x,y): x^2 + y^2 - xy \leq 2\}$.

22.Show that $\int_0^1 \int_0^1 \frac{1}{1+xy} dxdy = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n^2}$

【2023-2024 Final】

19. Let $x+y-\sqrt{6}z=0$ and $x+y+\sqrt{6}z=0$ be two planes' equation.
(1) Find the smallest positive angle between these two planes.
(2) Find the parametric equations of the line where these two planes intersect.

20. Find the global maximum and global minimum for the function $f(z,y)=x^2+y^2$over the closed and bounded set $S=\{(r,y):x^2+y^2-xy\leq 9\}$.

21. Let G be a hemisphere with the equation $x^2+y^2+z^2=1, z\geq 0$.
(1) Find the volume of the solid bounded above by G and below by the plane $z=h(0<h<1)$.
(2) Find the surface area of the part of G above the plane $z=h$.

22. Given series $1-\frac{1}{4}+\frac{1}{7}-\frac{1}{10}+\frac{1}{13}+\cdots.$
(1) Show that this series converges.
(2)Find the sum of this series.

【2021-2022 Final】

19.

【Solution】

$$(u \cdot v)v = 2v$$

【辅助理解】

先计算括号内点乘，再计算叉乘。

20.

【Solution】

$$\sum_{n=0}^{\infty} \frac{3^n + 2^n}{4^n} = \sum_{n=0}^{\infty} (\frac{3}{4})^n + (\frac{1}{2})^n = \frac{1}{1-\frac{3}{4}} + \frac{1}{1-\frac{1}{2}}$$

【辅助理解】

把原式子拆成两个几何级数的乘机，几何级数的 S 套公式即可，注意区分 a 和 r 的具体数值。

21.

【Solution】

$\int_{-1}^{1}\int_{0}^{3}\left(x^5y + xe^{xy}\right)dydx = \int_{-1}^{1}[\frac{1}{2}x^5y^2 + e^{xy}]_0^3\,dx = [\frac{3}{4}x^6 + \frac{e^{3x}}{3} - x]_{-1}^{1}$

【辅助理解】

按照积分顺序带入求解即可。

22.

【Solution】

Let $f = \ln(x+y+z) - x + 2y + 3z$.

$\frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial y} = 0 \rightarrow \frac{\partial z}{\partial y} = -\frac{\frac{\partial f}{\partial y}}{\frac{\partial f}{\partial z}} = -\frac{\frac{1}{x+y+z}-2}{\frac{1}{x+y+z}-3}$ when（$(x,y,z) = (-1,5,-3)$）

【辅助理解】

含有隐函数的多元函数求导，应套用公式$\frac{\partial z}{\partial y} = -\frac{\frac{\partial f}{\partial y}}{\frac{\partial f}{\partial z}}$求解。

23.

【Solution】

$f_x = yx^{y-1}, f_y = x^y \ln x$, $dx = \frac{1}{10}, dy = -\frac{1}{5}$

$\therefore dz = dxf_x + dyf_y$

【辅助理解】

此处考察全微分公式的运用，即 $dz = dxf_x + dyf_y$，dx 和 dy 的数值已经给出只需

求出 x 和 y 的偏导数再代入公式即可。

24.

【Solution】

Firstly, find the critical points in the interior of the region S

$$f(x,y) = 4x - 3y + 2$$

$$\frac{\partial f}{\partial x} = 4 \neq 0, \frac{\partial f}{\partial y} = -3 \neq 0$$

So there are no critical points in the interior of the region $S = \{(x,y): 2x^2 + 3y^2 < 1\}$

Then, use Lagrange multipliers to find the extreme values on the boundary of the region S

The boundary of the region S is given by $g(x,y) = 2x^2 + 3y^2 - 1 = 0$

The Lagrange equations are $\nabla f = \lambda \nabla g$,where $\nabla f = < \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} > = < -4,3 >, \nabla g = < \frac{\partial g}{\partial x}, \frac{\partial g}{\partial y} > = < 4x, 6y >$,So we have the following system of equations:

$$\begin{cases} 4 = 4\lambda x \\ -3 = 6\lambda y \\ 2x^2 + 3y^2 - 1 = 0 \end{cases}$$

We get $\begin{cases} x_1 = \frac{2}{\sqrt{11}} \\ y_1 = -\frac{1}{\sqrt{11}} \end{cases}$ and $\begin{cases} x_2 = -\frac{2}{\sqrt{11}} \\ y_2 = \frac{1}{\sqrt{11}} \end{cases}$

$$f(x_1, y_1) =$$
$$f(x_2, y_2) =$$

Since there are no critical points in the interior of the region S, the maximum value and the minimum value can be found.

【辅助理解】

考虑多元连续函数在有界的区域上的最大最小值时，应该考虑函数在 S 内部可能的极值点和在边界上可能的极值点。讨论 S 内部 $2x^2 + 3y^2 < 1$ 的极值点需要考虑驻点即偏导数为 0 的点。讨论 S 边界 $2x^2 + 3y^2 - 1 = 0$ 上的点可以用拉格朗日乘数求条件价值的方法找出，所有店带入 f 后再比较大小，找出最值。

25.

【Solution】

(1) $3x^2 - 7x + 2 = (3x - 1)(x - 2)$

We assume the partial fraction decomposition $\frac{x+3}{3x^2-7x+2} = \frac{A}{(3x-1)} + \frac{B}{(x-2)}$

Then calculate.

(2) $f(x) = \frac{x+3}{3x^2-7x+2} = \frac{-2}{(3x-1)} + \frac{1}{(x-2)}$

If t=x-1, $\frac{-2}{(3x-1)}=\frac{-1}{1-(-\frac{3}{2}t)}=-\sum_{n=0}^{\infty}\left(-\frac{3}{2}t\right)^n, \frac{1}{(x-2)}=\frac{1}{t-1}=-\sum_{n=0}^{\infty}t^n$

$$f(x)=\sum_{n=0}^{\infty}\left(-\left(-\frac{3}{2}\right)^n-1\right)t^n=\sum_{n=0}^{\infty}\left(-\left(-\frac{3}{2}\right)^n-1\right)(x-1)^n$$

The series converges when $\left|-\frac{3}{2}(x-1)\right|<1, |x-1|<1$

At $x=\frac{3}{5}/\frac{1}{3}$,the series diverges due to the n-th term test.

Hence, we can find the convergence set.

【辅助理解】

(1) 部分分式分解，把分母因式分解，分子设特殊值，通分后解方程即可。

(2) 由于两个部分均可以用几何级数和来替换，所以可以直接表示为几何级数的展开式。所以本题不用泰勒公式的一般项通式求解，无需复杂的求导运算。收敛域判断则根据几何级数收敛性判断公式即可，端点要具体带入，判断是否取舍，本题在端点都发散，所以是开区间。

26. 【Solution】

(1) The distance formula $u(x,y,z)$ is $\sqrt{(x-0)^2+(y-0)^2+(z-1)^2}$

Then simplify it in the spherical coordinates by using

$$x=\rho\sin\emptyset\cos\theta, y=\rho\sin\emptyset\sin\theta, z=\rho\cos\emptyset$$

(2) The volume-element in spherical coordinates is $dV=\rho^2\sin\phi\, d\emptyset d\rho d\theta$

The region B in in spherical coordinates is defined by

$0\le\rho\le 1, 0\le\emptyset\le\pi, 0\le\theta\le 2\pi$

From(1) we have $u(\rho,\emptyset,\theta)=\sqrt{\rho^2-2\rho\sin\emptyset+1}$

We can write the triple-integral $\iiint_B u(x,y,z)dV$.

(3)Find the volume of $V_B=\frac{4}{3}\pi r^3=\frac{4}{3}\pi$,then evaluate $\iiint_B u(x,y,z)dV$ in the order $d\emptyset d\rho d\theta$ .Substitute $\iiint_B u(x,y,z)dV$ and $\frac{4}{3}\pi$ into $m=\frac{\iiint_B u(x,y,z)dV}{V_B}$,then find the answer,

【辅助理解】

(1) 空间坐标系中点与点之间的距离公式代入在用球面坐标系换元。

(2) 找到所替换的三个未知量$\rho,\emptyset,\theta$的上下限，再替换即可。

(3) 算出第二问所写的三重积分算出答案，再除以体积V_B即可。

【2022-2023 Final】

19.

【Solution】

(1) $\begin{cases} z = x^2 + y^2 (z \geq 0) \\ x^2 + y^2 + z^2 = 2. \end{cases}$ So z=1.The projection curve of the intersection line on the xy plane is a circle with the origin as the center and radius of 1.

Since $Z_x =- 2x, Z_y =- 2y$,the integral of the square root of $(1 + {Z_x}^2 + {Z_y}^2)$ over the surface is given by $\iint \sqrt{1 + {Z_x}^2 + {Z_y}^2}\ dA$

Use polar coordinates to replace with $\{(r, \theta: 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi\}$

$$\int_0^{2\pi} \int_0^1 r\sqrt{1 + 4r^2}\ drd\theta$$

Then calculate….

(2) Let S be the set of points defined by the following inequalities:

$$S = \{(r, \theta, z) | 0 \leq \theta \leq 2\pi, 0 \leq r \leq 1, \sqrt{2 - r^2} \leq z \leq r^2\}$$

The volume V of the region S is given by the triple integral

$$V = \int_0^{2\pi} \int_0^1 \int_{\sqrt{2-r^2}}^{r^2} r^2 dzdrd\theta$$

Then calculate….

【辅助理解】

(1) 求抛物面再求内部分的面积，被积函数需要带入公式$\sqrt{1 + {Z_x}^2 + {Z_y}^2}$，再用极坐标换元简化，求出上下限算积分即可。

(2) 求被围成面积的体积，被积函数为 1。根据题意，围成的立体图形位于抛物面之下，球面之上，想象成类似碗装的空间区域（抛物面是碗底，球面是碗口）。因为题目抛物面方程为 $z = x^2 + y^2$，所以与 xy 平面平行的面交抛物面为圆形，整个立体图形可以放在柱面坐标系中换元。换元简化，求出上下限算积分即可。

20.

【Solution】

(1) The gradient of $F(x) = x^2 + y^2 - 1$ is

$\nabla F(x, y, z) = (2x, 2y, 0))$

The gradient of $G(x) = (x - 1)^2 + (y - 1)^2 + (z - 1)^2 - 1$ is

$\nabla G(x, y, z) = (2(x - 1), 2(y - 1), 2(z - 1))$

At the point (0,1,1),the gradients are

$\nabla F(x, y, z) = (0,2,0), \nabla G(x, y, z) = (-2,0,0)$

The direction vector of tangent line equation of the intersection curve is the cross product of the gradients at (0,1,1) is $\nabla F \times \nabla G = (0,0,4)$

So the parametric equation of the tangent line can be written.

(2) The gradient of $F(x) = x^2 + y^2 - 1$ in (x_0, y_0, z_0) is

$\nabla F(x, y, z) = (2x_0, 2y_0, 0)$

So the tangent plane equation of the cylinder in (x_0, y_0, z_0) is

$2x_0(x - x_0) + 2y_0(y - y_0) = 0$

$2x_0x + 2y_0y = 2{x_0}^2 + 2{y_0}^2 = 2$

$x_0x + y_0y = 1$

The tangent plane equation of the cylinder is similar to the tangent plane equation of the sphere. From the tangency condition, the distance from a point on the sphere to the plane is equal to the radius of the sphere.

The center of the sphere is (1,1,1),r=1,plane equation is $x_0x + y_0y = 1$

So $\frac{|x_0+y_0-1|}{\sqrt{{x_0}^2+{y_0}^2}} = 1$，${x_0}^2 + {y_0}^2 = 1$

$$\begin{cases} x_0 = -\frac{\sqrt{2}}{2} \\ y_0 = \frac{\sqrt{2}}{2} \end{cases} \begin{cases} x_0 = \frac{\sqrt{2}}{2} \\ y_0 = -\frac{\sqrt{2}}{2} \end{cases}$$

The equation of plane tangent to both the sphere and the cylinder can be written.

【辅助理解】

（1） 两个曲面的相交曲线在某点的切线位于两个曲面再该点的切平面内，而一点处的梯度向量和在该点切平面的法向量是平行关系（此处不予以证明）。所以相交曲线的切线的方向向量与两个曲面在（0,1,1）处的切平面法向量垂直，也与$\nabla F, \nabla G$ 都垂直。所以可以用梯度的叉乘来确定切线的方向向量。再用参数方程表示空间内的直线方程。

（2） 从特殊曲面的的切平面下手可以简化所设的平面方程，本题先求出圆柱的切平面，然后从题目条件入手，已知特殊点和特殊点面距离，结合已点到平面距离方程，求出平面方程。

21.

【Solution】

First, find the first order partial derivatives $f(x, y) = x^4 + y^4 - 4xy + 1$,then get the possible critical points:

$$\frac{\partial f}{\partial x} = 4x^3 - 4y = 0, \frac{\partial f}{\partial y} = 4y^3 - 4x = 0$$

$$\begin{cases} x = 0\ y = 0\ f(0,0) = 1 \\ x = 1\ y = 1\ f(1,1) = -1 \\ x = -1\ y = -1\ f(-1, -1) = -1 \end{cases}$$

Next, check for critical points on the boundary. Since S is a closed and bounded set, we need to consider the boundary condition$x^2 + y^2 - xy = 2$.

Use Lagrange multipliers to find the extreme values on the boundary of the

region S:

$$\begin{cases} 4x^3 - 4y = \lambda(2x - y) \\ 4y^3 - 4x = \lambda(2y - x) \\ x^2 + y^2 - xy = 2 \end{cases}$$

We can find $(\sqrt{2}, 0), (0, \sqrt{2}), (0, -\sqrt{2})(-\sqrt{2}, 0)$with f=5.

We can write the maximum and minimum.

【辅助理解】

依然是多元函数求最值问题，范围内部用求驻点的方式找到特殊点，边界用拉格朗日乘数法求出特殊点，通过比较求出最大最小值。

22.

【Solution】

$$\frac{1}{1+xy} = \sum_{n=1}^{\infty}(-xy)^n = \sum_{n=1}^{\infty}(-1)^n(xy)^n$$

$$\int_0^1\int_0^1 \frac{1}{1+xy}dxdy = \int_0^1\int_0^1 \sum_{n=1}^{\infty}(-1)^n(xy)^n\,dxdy = \sum_{n=1}^{\infty}(-1)^n\int_0^1\int_0^1 (xy)^n\,dxdy$$

$$= \sum_{n=1}^{\infty}(-1)^n\int_0^1 \frac{y^n}{n+1}dy = \sum_{n=1}^{\infty}(-1)^n\frac{1}{(n+1)^2}$$

【辅助理解】

右式可以写作几何级数的通式，再求积分即可推出和左式相等。

【2023-2024 Final】

19.

【Solution】

(1) Find the normal vectors of two planes.

We can get $(1,1,-\sqrt{6}), (1,1,\sqrt{6})$

The formula for the angle between two non-zero vectors is $\cos\theta = \frac{|\overrightarrow{n_1}\cdot\overrightarrow{n_2}|}{|\overrightarrow{n_1}||\overrightarrow{n_2}|}$

$|\overrightarrow{n_1}\cdot\overrightarrow{n_2}| = -4, |\overrightarrow{n_1}| = |\overrightarrow{n_2}| = 2\sqrt{2}$

So we can calculate θ with $\theta \in [0, \frac{\pi}{2}]$

(2) To find the line of intersection of the two planes, we can solve the systems of equations $\begin{cases} x + y - \sqrt{6}z = 0 \\ x + y + \sqrt{6}z = 0 \end{cases}$

We get y=-x

Then we can write parametric equations.

【辅助理解】

(1) 求两平面的最小夹角时，若已知两个平面方程，可以求出他们的法向量，法向量夹角与平面最小夹角互补，cos 值相同，可得出答案。

(2) 联立方程组可以得出交线方程，再写成参数方程形式即可。

20.

【Solution】

First, find the first order partial derivatives $f(x,y)=x^2+y^2$,then get the possible critical points:

$$\frac{\partial f}{\partial x}=2x=0,\frac{\partial f}{\partial y}=2y=0$$

We get $x=0\ y=0\ f(0,0)=0$ satisfying the region S={(r,y):x²+y²-xy≤9}.

Next, check for critical points on the boundary. Since S is a closed and bounded set, we need to consider the boundary condition$x^2+y^2-xy=9$.

Use Lagrange multipliers to find the extreme values on the boundary of the region S:

$$\begin{cases} 2x=\lambda(2x-y) \\ 2y=\lambda(2y-x) \\ x^2+y^2-xy=9 \end{cases}$$

We can find $(3,-3),(-3,-3),(\sqrt{3},-\sqrt{3})(-\sqrt{3},\sqrt{3})$with f=18/6.

We can write the maximum and minimum.

【辅助理解】

依然是多元函数求最值问题，范围内部用求驻点的方式找到特殊点，边界用拉格朗日乘数法求出特殊点，通过比较求出最大最小值。

21.

【Solution】

(1) First, convert the equation of sphere x²+y²+z²=1 into cylindrical-coordinates,$x=r\cos\theta,y=r\sin\theta,z=z$

So the equation of sphere becomes $r^2+z^2=1,z=\sqrt{1-r^2}(z\geq 0.)$

The intersection of the hemisphere x²+y²+z²=1, z≥0 and the plane z=h(0<h<1) gives $r=\sqrt{1-h^2}$

The volume V in cylindrical-coordinates is $\int_0^{2\pi}\int_0^{\sqrt{1-h^2}}\int_h^{\sqrt{1-r^2}} r\,dzdrd\theta$

Then calculate….

(2) The surface area S of z=f(x,y) is given by S=$\iint\sqrt{1+{Z_x}^2+{Z_y}^2}\ dA$

$$Z_x=-\frac{x}{\sqrt{1-x^2-y^2}},Z_y=-\frac{y}{\sqrt{1-x^2-y^2}}$$

Then $\sqrt{1+{Z_x}^2+{Z_y}^2}=\frac{1}{\sqrt{1-x^2-y^2}}$

Using polar coordinates, the region in the xy-plane is $0\le r\le\sqrt{1-h^2}, 0\le\theta\le 2\pi$.Then $S=\int_0^{2\pi}\int_0^{\sqrt{1-h^2}}\frac{1}{\sqrt{1-r^2}}\cdot r\,drd\theta$

Calculate…..

【辅助理解】

(1) 半球体具有绕着 z 轴的旋转对称性，而柱面坐标系可以很好利用这种对称性，与 z 轴垂直的平面可以用 r 和θ表示，被 z=h 所截后的图形求积分限相对简单。题意是求体积，所以原被积函数是 1，用柱面坐标系换元再积分即可。

(2) 由公式可得，求表面积的被积函数是$\sqrt{1+{Z_x}^2+{Z_y}^2}$，被积函数含有 x 与 y 的平方和，方便用极坐标系换元再积分。

22.

【Solution】

(1)The given series is an alternating-series$\sum_{n=0}^{\infty}(-1)^n\frac{1}{3n+1}$ with $a_n=\frac{1}{3n+1}$

$a_n>0, a_{n+1}\le a_n$for all n.$\lim_{n\to\infty}a_n=0$

This series converges.

(2)We consider a power-series$S(x)=\sum_{n=0}^{\infty}(-1)^n\frac{x^{3n+1}}{3n+1}, S'(x)=\sum_{n=0}^{\infty}(-1)^n x^{3n}$.

$\sum_{n=0}^{\infty}(-1)^n x^{3n}$is a geometric series with$r=-x^3, a=1$

So$\sum_{n=0}^{\infty}(-1)^n x^{3n}=\frac{1}{1+x^3}(|x^3|<1，|x|<1)$

$$S(x)=\int\frac{1}{1+x^3}dx=\frac{1}{3}\ln(x+1)-\frac{1}{6}\ln(x^2-x+1)+\frac{1}{3}\arctan\left(\frac{2x-1}{\sqrt{3}}\right)+C$$

X=1, $S(x)=\sum_{n=0}^{\infty}(-1)^n\frac{x^{3n+1}}{3n+1}=\sum_{n=0}^{\infty}(-1)^n\frac{1}{3n+1}$ converges .

$S=\frac{1}{3}\ln(1+1)-\frac{1}{6}\ln(1^2-1+1)+\frac{1}{3}\arctan\left(\frac{2-1}{\sqrt{3}}\right)$

We can write the sum of this series.

【辅助理解】

(1) 写出交错级数的通项公式，再判定即可。

(2) 构造 $S(x)=\sum_{n=0}^{\infty}(-1)^n\frac{x^{3n+1}}{3n+1}$幂级级数简化求和，通过对其求导得出几何级数$\sum_{n=0}^{\infty}(-1)^n x^{3n}$，能方便的求出函数和$\frac{1}{1+x^3}$，再通过积分求 S（x）求和公式，令 x=1 即可（S（x）的收敛区间在此不进行计算，答案是（-1,1],因为 $S(1)=\sum_{n=0}^{\infty}(-1)^n\frac{x^{3n+1}}{3n+1}=\sum_{n=0}^{\infty}(-1)^n\frac{1}{3n+1}$,即 S(x)级数在 x=1 端点处收敛,所以 1 可以带入）

MTH008 Resit Exams

【2021-2022 Resit】

17. Let $u = 3i + 2j - 2k, v = -j + 4k.$ then $u \times (3u + v) =$ _____

18. Evaluate $\int_1^3 \int_0^1 \frac{x+y}{1+x^2} dxdy =$ _____.

19.If $g(x, y) = \sin(xy^2)$, then $\frac{\partial^2 g}{\partial x \partial y} =$ _____

20.Let $z = \sqrt{x^2 + y}, x = 2s + 3t, y = st.$ Then $\frac{\partial z}{\partial t}|_{s=-2,t=3} =$ _____

23.For $A(-1,0,1)B(10,-1,6)C(5,0,-3)D(5,-2,1)$,plane π_1 passes through A, B, C, plane π_2 passes through A, B, D.
(1)Find the equations of the planes π_1 and π_2.
(2)Suppose that sphere S centered at $(2,1,t)$ is tangent to both π_1 and π_2.Find the value(s) of t.

【2022-2023 Resit】

19.Consider a sphere with equation $x^2 + y^2 + z^2 = 2$ and a half cone with the equation $z = \sqrt{x^2 + y^2}$.

(1) Find the area of the part of cone, which is inside the sphere.

(2) Find the volume of the solid, which is bounded above by the sphere and below by the cone.

20.Consider a one sheet hyperboloid with equation $x^2 + y^2 - z^2 = 1$.

(1) Given a cylinder $x^2 + y^2 = 4$, find the tangent line equation of the intersection curve at point $(\sqrt{2}, \sqrt{2}, \sqrt{3})$.

(2) Find the tangent plane of the hyperboloid, which is parallel to x-axis and through point $\left(0, \frac{1}{2}, 0\right)$.

21.Find the extreme values of $f(x, y) = \cos x + \cos y + \cos(x + y)$.

22.Show series $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots$ convergent, and find the sum of the series.

【2023-2024 Resit】

19.Let $x + y - z = 0$ and $x + y + 2z = 0$ be two plane's equation.

(1) Find the angle between these two planes.

(2) Find the equations of the line where these two planes intersect.

20. Find the global maximum and global minimum for the function $f(x, y) = 4 + xy - x^2 - y^2$ over the closed and bounded set $S = \{(x, y): x^2 + y^2 \leq 2\}$

21.Let G be an elliptic paraboloid with the equation $z = x^2 + y^2$.

(1) Find the volume of the solid enclosed by G and z = h $(0 < h < 1)$

(2) Find the surface area of the part of G under z = h.

22.Given series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} + \ldots$.

(1) Show that this series converges.

(2) Find the sum of this series.

【2021-2022 Resit】

17.

【Solution】

Firstly, let's figure out $3\mathbf{u}+\mathbf{v}=\langle 9,5,-2\rangle$, and then figure out $\mathbf{u}\times(3\mathbf{u}+\mathbf{v})$ by determinant.

18.

【Solution】

【辅助理解】：注意到$(\arctan x)'=\frac{1}{1+x^2}$，我们不妨使用分部积分$\int udv=uv-\int vdu$，对 x 积分时将 y 视为常数，并且可以拆分$\frac{x+y}{1+x^2}=\frac{x}{1+x^2}+\frac{y}{1+x^2}$再去积分，将会更有思路。

By formula of integration by parts, $u=x+y,\ v=\arctan x$

$$\int_1^3\int_0^1\frac{x+y}{1+x^2}dxdy=\int_1^3\int_0^1(x+y)d\arctan xdy$$

$$=\int_1^3([(x+y)\arctan x]_0^1-\int_0^1\arctan xdx)dy$$

Then the result can be obtained by iterated integral.

$$\int_1^3(\frac{1}{2}\ln 2+\frac{\pi}{4}y)dy$$

19.

【Solution】

Second order mixed derivative $\frac{\partial^2 g}{\partial x\partial y}$, we take the partial derivate with respect to x, and then we take the partial derivative with respect to y.

20.

【Solution】

Chain rule $\frac{\partial z}{\partial t}|_{s=-2,t=3}=\frac{\partial z}{\partial x}\frac{dx}{dt}+\frac{\partial z}{\partial y}\frac{dy}{dt}$

23.

【Solution】

(1) We have $\overrightarrow{AB}=(11,1,-7),\overrightarrow{AC}=(6,0,-4)$

∴ The normal vector for π_1:$n_1=\overrightarrow{AB}\times\overrightarrow{AC}=(-4,2,-6)$

Similarly, $\overrightarrow{AD}=(6,-2,0)$

∴ The normal vector for π_2:$n_2 = \overrightarrow{AB} \times \overrightarrow{AD} = (-14, -42, -28)$

The equations of π_1 and π_2 can be wrote.

(2) Since the sphere S is tangent to both π_1 and π_2 ,the distance from the center of S to the plane are equal. By the distance formula

$$\frac{|2 \cdot 2 - 1 + 3t - 1|}{\sqrt{2^2 + (-1)^2 + 3^2}} = \frac{|2 + 3 \cdot 1 + 2t - 1|}{\sqrt{1^2 + 3^2 + 2^2}}$$

The value(s) of t can be found.

【辅助理解】

（1）已知平面上的三点，可写出两个平面上的向量，用叉乘求出法向量。再用点法式方程求出平面方程。即已知法向量（a,b,c）和面上一点（d,e,f）,平面方程为：a(x-d)+b(y-e)+c(z-f)=0

（2）球体与两个面相切，那么球心与两个面的距离一定相等且等于球的半径。用点到平面距离公式求解。

【2022-2023 Resit】

19.

【Solution】

(1)【辅助理解】：第一问求面积，找到 $z = f(x, y)$再套用曲面表面积公式即可得出答案，注意确定积分区域（本题为圆心为原点，半径为 1 的圆），被积函数或积分区域出现$x^2 + y^2$的形式，可优先考虑极坐标换元（注意此时雅可比行列式的绝对值为 r）

Adjoining two equations of the sphere and the cone,

$\begin{cases} x^2 + y^2 + z^2 = 2 \\ z = \sqrt{x^2 + y^2} \end{cases}$, we can find domain of integration $0 \le x^2 + y^2 \le 1$

Additionally, $z = \sqrt{x^2 + y^2}$, $f_x = x(x^2 + y^2)^{-\frac{1}{2}}, f_y = y(x^2 + y^2)^{-\frac{1}{2}}$

Let $x = r\cos\theta,\ y = r\sin\theta$

$$A = \int_0^{2\pi} \int_0^1 \sqrt{(f_x)^2 + (f_y)^2 + 1}\ r dr d\theta$$

(2)【辅助理解】：第二问求体积，通过 $V = \iiint 1\, dV$ 三重积分求出答案，仍需优先确定积分区域，本题体积由一个圆锥和一个球缺组成，也推荐使用球坐标去解决，并且在球坐标下的式子为$\int_0^{\frac{\pi}{4}} \int_0^{2\pi} \int_0^{\sqrt{2}} r^2 \sin\varphi\, dr d\theta d\varphi$.

Triple integrals in cylindrical coordinate, let $x = r\cos\theta, y = r\sin\theta, z = z$

Obviously, the upper limit of the integral is $\sqrt{2 - x^2 - y^2}$ and the lower integral limit of z is $\sqrt{x^2 + y^2}$.

$$V = \int_0^{2\pi}\int_0^1\int_r^{\sqrt{2-r^2}} r\,dzdrd\theta$$

20.

【Solution】

(1) 【辅助理解】:

①法一：先验证点在交线上，再通过对两个曲面在该点处的切平面的法向量叉乘，求出切线的方向向量。

Obviously, $(\sqrt{2},\sqrt{2},\sqrt{3})$ is on both the cylinder and the circle.

Let **F** (x, y, z) = $x^2+y^2-z^2$, **G** (x, y, z) = x^2+y^2, then figure out

n = ∇**G** = ⟨2x,2y,0⟩, **m** = $\nabla\mathbf{F} = \langle 2x, 2y, -2x\rangle$

The direction vector of tangent line is T = $\mathbf{n}\times\mathbf{m}$
Finally, we can write tangent line at the point easily.

②法二：先联立求出两个曲面的交线方程，再将这个曲线参数化用向量值函数表示，求导带入即可得该点处的切线方向向量。
Firstly, we simultaneous equations of two surfaces to determine intersection equation $\begin{cases} x^2+y^2=4 \\ z=\pm\sqrt{3}\end{cases}$, and we notice $(\sqrt{2},\sqrt{2},\sqrt{3})$ is on the intersection. Secondly, the intersection can be written $\mathbf{r}(\alpha) = 2\cos\alpha\, i + 2\sin\alpha\, j + \sqrt{3}k$, then $\alpha = \frac{\pi}{4}$, and we put $\alpha = \pi/4$ into $\mathbf{r}'(\alpha)$ to find the direction vector of tangent line at the point.
Finally, we can write tangent line at the point easily.

(2) 【辅助理解】:

梯度与切平面垂直，即与 x 轴垂直, 再设平面点法式方程，最后将经过的点带入平面方程求出切点即可。

$\nabla\mathbf{F}\cdot\langle 1,0,0\rangle = 0$, x = 0

The tangent point (0, b, c) satisfies the $x^2+y^2-z^2=1$
The tangent plane: $2b(y-b)-2c(z-c)=0$

The plane passes through (0, $\frac{1}{2}$, 0), then $b=2, c=\pm\sqrt{3}$

Thus, the equation of tangent plane can be written easily.

21.

【Solution】

【辅助理解】：

法①：通过判别式判断极值点, 注意三角函数的多解

Firstly, we make gradient is equal to zero in order to find stationary points

$\frac{\partial f}{\partial x} =- \sin x - \sin(x+y)$, $\frac{\partial f}{\partial y} =- \sin y - \sin(x+y)$

Let $\nabla f(x,y) = 0$, then x = y or x = $\pi - y$

When x = y, $x = k\pi, \frac{2\pi}{3} + 2k\pi, \frac{4\pi}{3} + 2k\pi$

When x = $\pi - y$, $x = k\pi, y = \pi - k\pi$

Secondly, we write the second partial derivative.

$f_{xx} =- \cos x - \cos(x+y)$, $f_{yy} =- \cos y - \cos(x+y)$, $f_{xy} =- \cos(x+y)$

Lastly, the extreme point is judged by the Hessian matrix.

$D = AC - B^2 > 0, A < 0,$ is extreme maximum.

$D = AC - B^2 > 0, A > 0,$ is extreme minimum.

$D = AC - B^2 < 0,$ is saddle point

法②：利用 xy 轮换对称性，对函数降维，变成一元函数求极值问题

Firstly, we notice the function is symmetric with respect to y = x.

Then, let y = x, $f(x,x) = 2\cos x + \cos 2x = 2\cos x + 2\cos^2 x - 1$

Let $z = \cos x \in [-1, 1]$, $f(z) = 2z + 2z^2 - 1$

Let $f'(z) = 0$ to find extreme points, and we need to examine boundary points when $z = \pm 1$

Actually, we also should verify the function value is larger or smaller when $x \neq y$, so we can let $y =- x$ and find the range of f is also in the interval.

22.

【Solution】

【辅助理解】：不难发现这是一个交错级数，那么我们写出它的通项，通过莱布尼兹判别法判断其敛散性。求常数项级数的和，可将目标级数与常见函数（如指数，三角，对数）的泰勒展开式对比去联系起来。观察题目所给的级数，我们发现它的每一项正好对应着 arctanx 泰勒展开后前面的系数，换言之，这个级数就是 arctanx 在 $x = 1$ 处的泰勒展开（答案即是 arctan1）。但是你可能对泰勒级数没有那么敏感，我们应该怎么做呢？另一种思路就是利用幂函数的和函数，对它进行逐项积分、微分等操作，最后再令 $x = 1$ 即可。

Consider the alternating series: $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{2n-1}$

The sequence $\frac{1}{2n-1}$ is strictly decreasing since $\frac{1}{2n-1} > \frac{1}{2n+1}$ for all $n \geq 1$.

And $\lim_{n\to\infty} \frac{1}{2n-1} = 0$, then the series converges by Leibniz Test.

Construct the power series: $S(x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n-1}}{2n-1}$.

Differentiate the power series: $\sum_{n=1}^{\infty}(-1)^{n-1}x^{2n-2}=\sum_{n=1}^{\infty}(-x^2)^{n-1}=\frac{1}{1+x^2}$

Integrate to Recover S(x): $\int_0^x \frac{1}{1+x^2}dx=\arctan(x)$

Finally, let $x=1$, then it is the sum of the series.

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19.

【Solution】

(1) 【辅助理解】：求平面之间的夹角转化为求法向量之间的夹角

$\mathbf{n}=\langle 1,1,-1\rangle$, $\mathbf{m}=\langle 1,1,2\rangle$，figure out $\cos\langle \mathbf{n},\mathbf{m}\rangle$

Thus, the angle between these two planes is found.

(2)【辅助理解】:

叉乘两个法向量求出交线的方向向量，再求两平面任意一个交点

$\mathbf{s}=\mathbf{n}\times\mathbf{m}$

Let x = 0 , $\begin{cases} y-z=0 \\ y+2z=0 \end{cases}$, then the intersect point is $(0,0,0)$

Thus, the line can be determined.

20.

【Solution】

【辅助理解】:

拉格朗日乘数法求有约束条件极值问题, 注意比较边界点等临界点

Let $L(x,y)=f(x,y)+\lambda\varphi(x,y)$, $\varphi(x,y)=x^2+y^2-2$

$\begin{cases} L_x=f_x+\lambda\varphi_x=0 \\ L_y=f_y+\lambda\varphi_y=0 \\ L_\lambda=\varphi=0 \end{cases}$, then $\begin{cases} y-2x+2\lambda x=0 \\ x-2y+2\lambda y=0 \\ x^2+y^2=2 \end{cases}$, $x=\pm 1, y=\pm 1$

Compare $f(1,1), f(-1,-1), f(1,-1), f(-1,1), f(0,\sqrt{2}), f(\sqrt{2},0), f(0,0)$

Thus, the global maximum can be written.

21.

【Solution】

(1) 【辅助理解】：第一问求体积，利用三重积分 $V=\iiint 1\,dV$ 求出答案，通过联立椭圆抛物面和平面方程，以找到在 xy 平面上的积分区域，再确定好 z 的积分上下限即可，需要注意的是本题难以使用球坐标求解（因为抛物面在球坐标下的表达式复杂，且积分限难以确定，而使用柱面坐标是极其方便的）

Adjoining two equations of the elliptic paraboloid and the plane

$\begin{cases} z=x^2+y^2 \\ z=h \end{cases}$, we can find domain of integration in xy-plane $0\le x^2+y^2\le h$

Let $x=r\cos\theta, y=r\sin\theta, z=z$ in cylindrical coordinate, the lower integral limit

of z is $x^2+y^2=r^2$

$$V=\int_0^{2\pi}\int_0^{\sqrt{h}}\int_{r^2}^{h} r\,dzdrd\theta$$

(2)【辅助理解】：第二问求面积，确定 z = f(x, y)，求出$\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$, 再套用曲面表面积公式即可得出答案

The domain of integration is $0\le x^2+y^2\le h$, let $x=r\cos\theta, y=r\sin\theta$

$$f_x=2x, f_y=2y$$

$$A=\int_0^{2\pi}\int_0^{\sqrt{h}}\sqrt{(f_x)^2+(f_y)^2+1}\,rdrd\theta=\int_0^{2\pi}\int_0^{\sqrt{h}}\sqrt{4r^2+1}\,rdrd\theta$$

In order to facilitate the solution, you can let $u=4r^2+1$.

22.

【Solution】

(1) 【辅助理解】：这显然是一个交错级数，那么我们可以通过莱布尼兹判别法去判断其收敛性（通项的绝对值在 n 趋于无穷大时必须单调递减且极限为 0）。

Consider the alternating series: $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n}$

Let $a_n=\frac{1}{n}$, then $\frac{1}{n}>\frac{1}{n+1}>0$ for all $n\ge 1$. The sequence is strictly decreasing.

Additionally, $\lim_{n\to\infty}\frac{1}{n}=0$. Thus, this series converges.

(2) 【辅助理解】：求常数项无穷级数的和，可考虑将目标级数与常见函数的泰勒展开式对比，通过观察，我们可以发现这个交错调和级数正好是 $\ln(1+x)$在 $x=1$ 处的泰勒展开式（答案即是 ln2）。如果对泰勒级数不够敏感，我们还可以构造幂级数(方便运算)，通过逐项微分、积分去计算和函数，最后令 $x=1$ 即可。

Construct the power series: $S(x)=\sum_{n=1}^{\infty}(-1)^{n+1}\frac{x^n}{n}$.

Differentiate the power series: $\sum_{n=1}^{\infty}(-1)^{n+1}x^{n-1}=\sum_{n=1}^{\infty}(-x)^{n-1}=\frac{1}{1+x}$.

Integrate to Recover S(x): $\int_0^x\frac{1}{1+x}dx=\ln(1+x)$.

Finally, let x = 1, then it is the sum of the series.